

Competing for Talent: Firms, Managers, and Social Networks

Isaac Hacamo

Indiana University

Kristoph Kleiner

Indiana University

Do social networks help firms recruit talented managers? In our setting, firms are randomly connected to prospective young managers through former employees. Under a discrete choice model, we find networks increase the likelihood firms hire high-ability managers, while having no effect on the hiring rate of low-ability managers. Effects are greatest for nonlocal firms, strong ties, and peers living in the same neighborhood. Survey evidence suggests social networks promote recruitment by providing information about firm fundamentals to potential applicants. Our results help rationalize why the majority of managers hold prior connections to the firm. (*JEL* G30, M12, M51, J24)

Received December 3, 2018; editorial decision December 24, 2020 by Editor Francesca Cornelli. Authors have furnished an Internet Appendix, which is available on the Oxford University Press Web site next to the link to the final published paper online.

The talent wars begin even before (MBA) classes do.

—Kelsey Gee, *The Wall Street Journal*¹

As managers hold significant influence on firm performance (Bertrand and Schoar 2003; Bennedsen, Pérez-González, and Wolfenzon 2016; Chang, Dasgupta, and Hilary 2010; Jenter, Matveyev, and Roth 2020),

We thank our editor Francesca Cornelli, two anonymous referees, Ken Ahern, Fernando Anjos, Iwan Barankay, Asaf Bernstein, Matt Billett, Inês Black, Diana Bonfim, Claire Celerier (discussant), Tom Chang, Tuugi Chulum (discussant), Jonathan Cohn, Andrew Ellul, Ted Fee (discussant), Miguel Ferreira, Cary Frydman, Will Gerken (discussant), Eitan Goldman, Nandini Gupta, Jerry Hoberg, Camelia Kuhnén, Chris Parsons, Francisco Queiró, Imran Rasul, Luc Renneboog (discussant), Nathan Seegert, Merih Sevilir, Kelly Shue, Scott Smart, Breno Schmidt (discussant), Christopher Stanton, Ed Van Wesep, Yongxiang Wang, Wenyu Wang, and Fernando Zapatero for very helpful suggestions. We also thank seminar participants at the Bank of Portugal, Indiana University, NOVA School of Business and Economics, and University of Southern California, and conference participants at the Cavalcade, Conference on Financial Economics and Accounting, EFA, Executive Compensation Conference at Rotterdam, Finance and Labor Working Group, FIRS, and the London Business School Summer Symposium for their comments. Send correspondence to Kristoph Kleiner, kleinerk@indiana.edu.

¹ Quotation comes from Gee (2017).

understanding how firms acquire the most productive managers is akin to understanding the sources of their value.² Recent evidence suggests that connections are decisive in CEO hiring as the vast majority of new CEOs have close connections to the hiring firm (Cziraki and Jenter 2020).³ One possible explanation for these findings is provided in the labor economics literature (Beaman and Magruder 2012; Dustmann et al. 2015; Burks et al. 2015; Pallais and Sands 2016; Brown, Setren, and Topa 2016; Heath 2018): firms benefit from the networks of their employees. However, evidence of these dynamics in the managerial labor market has been limited.⁴ Analyzing an experimental setting that randomly connects firms and managerial applicants, this paper demonstrates how social networks can improve a firm's recruitment of talented managers.

To conduct our analysis, we study the careers choices of young managers that graduated from the Master of Business Administration (MBA) program at Indiana University.⁵ As discussed in Bhagat, Bolton, and Subramanian (2010), nearly 40% of public firm CEOs hold an MBA degree; Indiana University is 1 of the 10 most common MBA programs these CEOs attend.⁶ Each year, various firms compete to hire new MBA graduates for managerial positions.⁷ The market to hire the best applicants is quite competitive; for example, recent IU graduates apply to only 7 jobs (among a sample of 80 firms), limiting the number of available applicants within each firm.⁸ By focusing on MBA graduates, we are able to clearly identify a population of managerial job applicants. Since the set of firms hiring newly graduated MBA students is highly concentrated, we can simultaneously identify a population of potential employers hiring managers

² Consistent with this view, recent work shows that firms with more productive human capital have higher market-to-book ratios, higher growth rates, and a lower cost of capital (Imrohoroglu and Tüzel 2014; Belo et al. 2017).

³ Cziraki and Jenter (2020) estimate the majority of outside CEO hires have prior connections to the firm's hiring board.

⁴ In fact, past evidence finds that managers with prior social connections to the board of directors face reduced monitoring, leading to worse firm outcomes (Fracassi and Tate 2012; Kramarz and Thesmar 2013). We differ from this literature by instead exploring connections between a managerial applicant and a former employee with the firm; as we focus on a different type of connection, our findings differ from this literature.

⁵ As a result, we follow a recent research evaluating the drivers of young managers' careers, including stock market conditions (Oyer 2008) or individual firms' stock returns (Bhole and Oyer 2014), gender (Bertrand, Goldin, and Katz 2010), risk aversion and optimism (Sapienza, Zingales, and Maestripieri 2009; Kaniel, Massey, and Robinson 2010), university ranking (Arcidiacono, Cooley, and Hussey 2008; Zimmerman 2016), previous industry experience (Kuhnen and Oyer 2016), job search intensity (Kuhnen 2017), and entrepreneurship (Lerner and Malmendier 2013).

⁶ In support of these findings, we estimate that 5% of the graduates in our sample eventually become a CEO and another 7% become a CFO.

⁷ MBA graduates are likely to become managers shortly after graduation. Confirming this prediction, we show that over 70% of the MBA graduates in our sample reach a middle manager position within 5 years of graduation.

⁸ This result is not unique to our data sample as Kuhnen (2017) estimates UNC MBA students apply to 13 positions when on the job market.

each year.⁹ The combined result is a unique setting to study both supply and demand in the managerial labor market.

To guide our hypothesis development, we extend contract theory to the study of social networks. According to our framework, social networks alleviate informational asymmetries regarding an employer's fundamentals, leading potential applicants to prefer the employer to alternative firms (Grossman and Hart 1983; Hart 1983; Azariadis 1983). Testing our hypotheses requires overcoming the standard endogeneity concerns inherent in social networks: (a) individuals often choose to interact with peers with similar unobservable characteristics and (b) peer groups often experience common unobservable shocks (Manski 1993). In our setting, this endogeneity is particularly problematic since individuals seek interactions with former employees of prospective firms. Fortunately, the MBA program at Indiana University (IU) provides a unique experimental setting since every entering MBA student is randomly assigned to a cohort and a team in their first year in the program. Individuals in the same cohort take the core MBA classes together, and those in the same team are required to work together on course projects and a large case study at the end of the first semester.¹⁰ We consider an individual to be exogenously connected to a firm if a former employee of that firm belongs to his or her team (or cohort).¹¹

Our analysis also requires an observable measure of ability estimated prior to social interaction. We follow a large literature in personnel psychology documenting general cognitive ability as the single best predictor of job performance (Hunter 1986; Ree and Earles 1992; Schmidt and Hunter 1998).¹² As such, we use scores on the Graduate Management Admission Test (GMAT), which evaluates an MBA applicant's analytical, writing, quantitative, verbal, and reading skills. Additionally, we use classroom performance in the first semester of the MBA program as a robustness measure.

To follow the career paths of managers, we obtain individual-level employment records from LinkedIn. This online business networking data include both employment and educational histories for workers as well as employer-level information detailing industry and headquarter locations. We merge these data with information about cohort and team assignments, admissions information, and class performance for each graduate, obtained from the MBA office at IU. We collect employment information for 95% of

⁹ In our sample, 10 firms hire nearly 25% of the graduating students.

¹⁰ An MBA class at IU has approximately 200 students per year during our analysis period. On average, a team and cohort comprise 4 and 6 students, respectively.

¹¹ Given that students do not choose their teams or cohorts, some individuals will be more connected to a given firm for reasons exogenous to their ability, effort, or interests.

¹² Recent papers have explored more direct measures of worker productivity (Burks et al. 2015; Pallais and Sands 2016; Brown, Setren, and Topa 2016); however, these papers are only able to focus on a single firm or industry. In comparison, our measures allow us to follow all workers, regardless of firm or industry.

all individuals graduating between 2003 and 2013, resulting in a final sample of nearly 2,200 young managers with an MBA degree from IU. We compile this data set into a panel structure where we observe the employment of each manager at the annual level for the first 5 years following graduation. In addition, we identify all places of employment for each individual prior to entering the Indiana University MBA program. With these data, we model employment outcomes under a conditional logit choice model that allows for both individual-specific attributes and alternative-specific characteristics as in McFadden (1973).

As we identify nearly 1,000 firms hiring IU graduates within 1 year after graduation, it is not computationally feasible to estimate a discrete choice model with all employers in our sample. Instead, we divide firms into smaller clusters, allowing us to simultaneously estimate a well-behaved discrete choice model as well as build an implicit ranking system for firms in our sample. We first focus our analysis on firms that send at least two students to the IU MBA program. Within this subset, we then cluster firms based on their hiring rate at the three top-ranked U.S. MBA programs and classify them into three categories: *top*-, *mid*-, and *bottom-tier* firms.¹³ Top-tier firms routinely recruit from top MBA programs, while bottom-tier firms rarely or never recruit from these elite programs. We estimate 19% of the sample previously worked at a top- or mid-tier firm prior to joining the MBA. In addition, 40% of graduates join one of these firms after graduation.¹⁴

We first evaluate the mean effect of former employees on the employment outcomes of managers. We estimate that having a team connection to a former employee of a top-tier (mid-tier) firm increases the odds of employment at the top-tier (mid-tier) firm within the first year of graduation by 3.5 (4.8) percentage points. We find similar effects on employment even 5 years after graduation, suggesting social connections induce permanent effects in the allocation of young managers to employers. In contrast, we find no effects when separating managers into placebo teams, suggesting our results are directly driven by social connections; similarly, we find no evidence that connections help managers gain employment at outside firms, suggesting a firm-specific network effect.

Second, we provide evidence consistent with the theory that social networks facilitate the flow of information. As we find network effects are primarily driven by former employees with longer job tenures, our findings suggest that connections with the largest amount of information exert the largest peer effects. In addition, having a team connection to a former employee of a bottom-tier firm increases the odds of employment at the bottom-tier firm within the first year of

¹³ We consider the Harvard, Chicago Booth, and Stanford University as the three most prestigious MBA programs in the United States. We provide further details of this classification system in the data section.

¹⁴ We provide details of the industry composition and employment size of these firms in Figures 2 and 3, respectively. For detailed information concerning each firm in this list, we refer the readers to Tables A1 and A2 in the Internet Appendix.

graduation by 5.6 percentage points. Assuming the asymmetry of information is larger for bottom-tier firms, these finds provide further support the results are driven by the exchange of information. Lastly, we find social influence is most important for geographically distant firms as well as employers that infrequently hire from IU; these are firms for which information is less readily available and less likely to have the resources to recruit through more formal channels.

Third, we compare the influence of strong and weak social ties. While influential work by Granovetter (1977) argues for the role of weak ties, Gee, Jones, and Burke (2017) finds empirical support that strong ties have greater influence on later employment. Our results support the findings of Gee, Jones, and Burke (2017): while the effect of an individual team member is 10 times larger the effect of an individual cohort member, the combined effect of all cohort members is greater than the combined team effect. In addition, splitting team members by residential location, we find network effects are greatest for team members living near one another. Similarly, we find evidence that team members of the same gender influence employment conditions with smaller effects from team members of the opposite gender.¹⁵

Fourth, we examine whether social networks influence the employment of high or low cognitive-ability managers. We split managers into terciles based on quantitative GMAT scores. We find limited evidence that social networks affect the employment outcomes of low-ability managers; in contrast, team connections increase the odds a given high-ability manager joins a mid- or top-tier firm by 2.9 to 3.5 percentage points. We confirm similar effects when using alternative measures of ability based on the combined GMAT score or first-semester grades in the MBA program. In addition, we show hires previously connected to the firm through a teammate have higher GMAT scores and better grades than unconnected managers hired by the same company. Last, we exploit variation in average cognitive ability across teams and find firms connected to a team with higher cognitive ability are more likely to hire a high-ability manager upon graduation.¹⁶

Fifth, we provide survey-level evidence that social networks affect recruitment outcomes by providing information about firm fundamentals to potential managers (Grossman and Hart 1983; Hart 1983; Azariadis 1983). Surveying recent MBA graduates and current MBA students from the classes of 2020 and 2021, we find social networks influence where students apply for internships between the first and second year and full-time offers at graduation. Having a connected team member increase the likelihood of applying for a

¹⁵ Our focus on gender is motivated by prior research in the referral literature and based on the argument of homophily, that is, that individuals sort into subgroups on their tendency to seek out those who are similar to themselves (Dustmann et al. 2015; Bandiera, Barankay, and Rasul 2009; Bertrand, Luttmer, and Mullainathan 2000).

¹⁶ We also provide evidence that connected managers stay longer at the firm relative to an unconnected manager at the same firm.

position at the firm by 19 percentage points. As the median MBA student applies to only 7 of the firms analyzed in our sample, the application decision can lead to significant impacts on employment outcomes. In addition, as 14% of workers report receiving a referral from a classmate, information may simultaneously flow from applicants and towards potential employers through networks (Montgomery 1991; Hensvik and Skans 2016).

Collectively, our results offer the first evidence that social networks improve managerial recruiting efforts. We recognize that in contrast to past research that focuses explicitly on CEOs, we analyze the market for mid-level managers: nearly 60% of our sample hold a managerial title within 1 year of graduation and over 70% within 5 years of graduation. Despite the differences, we believe our analysis of the young managerial labor market holds valuable lessons for the labor market of more senior managers. First, while the graduates in our sample are not yet executives, we estimate that 5% (7%) eventually reach the position of Chief Executive Officer (Chief Financial Officer). These findings are also backed by Bhagat, Bolton, and Subramanian (2010), who find Indiana University is one of the 10 most common MBA programs to graduate from among U.S. public firm CEOs. Second, as the vast majority of firms in the United States are quite small, many firms likely face constraints when recruiting managers because of informational frictions and insufficient resources.¹⁷ Third, even the labor market for public firm CEOs is relatively segmented.¹⁸ Because of this segmentation, social networks may continue to influence the decision to apply for positions and accept offers, leading to increased access to human capital.

1. Literature Review

Our paper provides contributions to two separate literatures: (1) the drivers and dynamics of managerial labor markets and (2) the effect of social networks on firm value more broadly. The former literature includes a range of recent papers including Yonker (2016), Nickerson (2017), Graham, Kim, and Kim (2020), and Frydman (2019). Closest to our work is Cziraki and Jenter (2020), who finds the majority of outside managerial hires are previously connected to the firm. We add to this literature by showing that firms can hire better managers through social networks, especially the networks of former employees. By directly surveying potential managers while on the job search, we also demonstrate

¹⁷ According to the 2017 census, over 89% of firms have under 20 employees, 9% have 20–99 employees, and 1.5% have 100–499 employees. In comparison, less than 0.5% of firms employ over 500 workers (<https://www.census.gov/data/tables/time-series/econ/susb/susb-historical.html>).

¹⁸ For example, 30% of public firm CEOs were born in the same state as the firm's headquarter, a rate five times the expected rate if labor markets were national (Yonker 2016). In addition, Zhao (2018) finds that over one-third of public firm CEO hires are from local firms. Similarly, Ma, Pan, and Stubben (2020) estimate that external hires of new CEOs (CFOs) are five (eight) times more likely to be from local firms than nonlocal firms.

that prospective managers may learn and choose firms based on information collected from current and former employees. In addition, our methodology provides a contribution by (a) simultaneously identifying both supply and demand in the managerial labor market and (b) introducing randomized closely defined peers to identify social network effects.

Second, our main results also contribute to a second literature analyzing the effect of social networks on firm value (Faccio 2006; Cai and Sevilir 2012; Engelberg, Gao, and Parsons 2012; Karolyi 2014; Haselmann, Schoenherr, and Vig 2018). Given the growing role of human capital as a driver of firm value (Zingales 2000; Belo et al. 2017; Agrawal, Hacamo, and Hu 2020), understanding how firms acquire human capital is akin to understanding how to maximize the value of the firm. From a methodological standpoint, we incorporate predicted measures of job performance based cognitive ability measured prior to interaction; we then document firms are more likely to hire a manager with greater cognitive ability when connected to these potential recruits through a former employee. In this way, we identify a novel asset to the firm: the networks of former employees. While investment in human capital is risky for firms due to worker mobility (Eisfeldt and Papanikolaou 2013; Lustig, Syverson, and Van Nieuwerburgh 2011; Donangelo 2014), our paper suggests firms may continue to benefit from their skilled workers even after turnover.

2. MBA Setting

Evaluating the causal effects of networks on managerial recruitment requires an experimental setting with four characteristics. First, researchers must observe the supply of potential applicants for managerial positions across a wide range of firms. This is difficult in most settings as it is nontrivial to identify workers that are both qualified for a given managerial position and looking for a job. Second, we must randomize connections between potential applicants and firms, allowing for both tightly defined connections and weakly defined connections. Without a randomized experiment, we cannot identify the causal effect of social networks. Third, we need *ex ante* measures of individual ability, and this measure should not be influenced by social networks. Traditionally, this has been an issue in the corporate governance literature, which has examined networks between managers and the board of directors. While directors are responsible for hiring a CEO, they also monitor the activities of the CEO and therefore influence performance. Fourth, to evaluate how social networks influence hiring outcomes we must observe the intermediate steps in the hiring process.

We argue the MBA labor market satisfies each of these four conditions. First, we observe both managerial labor demand (as a small number of firms are responsible for a significant share of MBA hiring each year) and labor supply (as the vast majority of students are applying for new positions and follow a

standard timetable).¹⁹ We confirm that the majority of the MBA graduates in our sample reach middle-manager status within few years after graduation.²⁰ We also confirm these same graduates reach executive positions later in their careers.²¹ Second, MBA students are assigned into large and small balanced peers groups in a randomly fashion, allowing us for causal inference after controlling for balancing characteristics—we detail this design in section 4. Third, MBA students are required to take GMAT (General Management Admission Test) when applying for MBA programs, which provides a measure of ability that is comparable across all individuals. Fourth, we can directly contact the survey MBA students, allowing us to observe intermediate steps in the job application process.

3. Conceptual Framework

3.1 Hypothesis development

Drawing from theory on implicit contracts under asymmetric information, (Grossman and Hart 1983; Hart 1983; Azariadis 1983) we argue social networks will increase the likelihood the potential applicant joins the firm.²² We provide the intuition of the theory below and the details of the theory in the Internet Appendix. Under perfect information, firms will tie wages to the productivity of the firm. However, when employees do not observe a firm's future productivity shocks, employment must instead be contracted on less efficient, but observable terms. This inefficiency leads to less preferable contracts with higher likelihood of future unemployment and lower offered wages. Assuming social networks can instead provide these productivity signals, connected firms can provide higher wages and lower likelihood of unemployment, making the firm more desirable from the perspective of the applicant. This theory then leads to our first testable hypothesis.

¹⁹ For confirmation of this argument, we estimate only 4.5% of the graduates in our sample return to a former employer within the first 5 years after graduation. In addition, based on self-reported survey responses, only 6% of current MBA students intend to return to a former employer.

²⁰ To confirm that graduates reach middle-manager status, we identify keywords in job titles used in managers' résumés. For simplicity, we define middle-manager status as encompassing six managerial titles: (1) Manager, (2) Director, (3) President or Vice President, (4) Executive, (5) Partner, and (6) Principal. According to Figure 1, we estimate over 55% of managers have a manager title; this percentage increases to over 70% in year 5. While the majority of titles include the word "Manager," we confirm a significant and rising fraction of higher-level titles over the 5-year period. For example, at year five, 14% of managers hold Director-level jobs, and 7% of job titles include the word "President" or "Vice President."

²¹ Based on the earlier cohorts in our sample, we estimate that 5% (7%) eventually reach the position of Chief Executive Officer (Chief Financial Officer).

²² Alternatively, social networks may influence recruitment outcomes by providing referrals to firms. There are two versions of these models. In the first version, valuable information is implicit in social networks because current and past employees are of known high-ability and interact with workers of similar quality (Montgomery 1991; Hensvik and Skans 2016). In the second version, the information is valuable because current and past workers have explicit or implicit incentives to be truthful when actively providing referrals (Pallais and Sands 2016; Heath 2018). Firms will then disproportionately make offers to candidates with referrals. The hypotheses developed below continue to hold under this alternative flow of information.

Hypothesis 1: Social connections increase the likelihood a given manager joins a given firm.

Under the first hypothesis, social networks provide value to firms. However, this does not mean the effects are homogenous across all firms. We hypothesize social networks are likely less relevant for well-known and local employers as potential applicants will have access to information through on-campus events and local alumni networks. These arguments then lead us to Hypothesis 2a.

Hypothesis 2a: The effect of social connections is greater for less-known firms.

The intensity of the relationship between social networks and employment outcomes may also depend on the tightness of the social connection between peers. On the one hand, early research by Granovetter (1977) argues that social structures comprise a collection of close friends (strong ties) and acquaintances (weak ties) and that weak ties are more likely to influence the rate of location of future employment. On the other hand, more recent work by Gee, Jones, and Burke (2017) examines social networking data from Facebook to document the opposite effect. They find that a user is more likely to join the same workplace as a given strong tie, but collectively weak ties are more important because they are more numerous.²³ This then leads to Hypothesis 2b.

Hypothesis 2b: The effect of social connections is greater for tight relationships.

Our study analyzes cognitive ability as a measure of predicted job performance. When cognitive ability is observable, we expect that the likelihood that firms make job offers is increasing with cognitive ability. We assume that if firms can observe ability, they make offers to applicants above a specific threshold that may depend on their overall productivity and profitability (Abowd, Kramarz, and Margolis 1999). Assuming students apply to only a small number of firms, employers are then limited in their access to students with high cognitive ability.²⁴ This then leads to Hypothesis 3.

Hypothesis 3: Social connections increase the likelihood a given high-ability manager joins a given firm relative to a low-ability manager.

The hypotheses above argue that social networks influence the employment outcomes of managers; however, the precise channel is less clear. Based on our theoretical model outlined in the Internet Appendix, we propose that managers learn about firm fundamentals through social networks, affecting the preferences of applicants. This then leads us to Hypothesis 4.

²³ They measure tie strength as the number of times two people interact in a year or as the number of mutual friends they share.

²⁴ In addition, assuming cognitive ability is not readily observable to firms, but is observable to social networks, referrals may provide critical information about cognitive ability. However, if former employees act under nepotistic motives (Goldberg 1982; Kramarz and Thesmar 2013; Kramarz and Skans 2014), social networks may actually hurt the firm.

Hypothesis 4: Managerial social connections affect employment outcomes by influencing the preferences of applicants.

4. Data

The contribution of this paper stems jointly from the random assignment of MBA students, data on long-term career outcomes at the individual level, and measures of cognitive ability. We first discuss and summarize our unique data sources: (a) the Indiana University Kelley School of Business MBA Program and (b) a large online business networking service that provides both employment histories and firm-level data.

4.1 Data sources

4.1.1 Kelley School of Business MBA. We analyze the employment outcomes of students from the Full-Time Indiana University Kelley School of Business MBA Program. Indiana University is located in Bloomington, Indiana, and has a total student population of 44,000. Among those students, roughly 400 are Full-Time MBA students. The Indiana University MBA program is ranked 21st according to the 2020 U.S. News and World Report Best Graduate Schools MBA Rankings and 25th according to the 2019 Bloomberg Businessweek MBA Rankings. Though the exact majors offered have evolved over time, students are generally able to major in Entrepreneurship, Finance, Management, Marketing, Operations, and Strategy. Following graduation, Indiana University is 1 of the 10 most common MBA programs to graduate from among public firm CEOs (Bhagat, Bolton, and Subramanian 2010).²⁵

4.1.2 Kelley School of Business MBA Admissions Information. To begin our analysis, we collect admissions data, including personal characteristics, such as GMAT scores, citizenship, gender, and ethnicity. We merge these admissions data with the team and cohort information. All entering full-time MBA students are assigned to 1 of 3 cohorts of approximately 60 students. Members of a cohort take the first semester classes together. In addition, students are assigned to a team within the cohort, which comprises four or five students. Members of a team compete in two different case competitions that compose part of their final grades for the semester. In addition, teams work together on group homework assignments.

The assignment process for the Kelley School of Business MBA Program attempts to maximize diversity across both cohorts and teams and is similar to the method used at Harvard Business School (as discussed in Shue 2013) and the University of Michigan (as discussed in Ahern, Duchin, and Shumway

²⁵ Other MBA programs in the top 10 are Harvard, Stanford, University of Pennsylvania, Northwestern, University of Chicago, Columbia, Dartmouth, NYU, and University of Virginia. Overall, nearly 40% of public firm CEOs hold an MBA degree.

2014).²⁶ Students entering prior to 2009 were assigned to their cohort/team by maximizing diversity across four characteristics: gender, race (for domestic students), citizenship (classified as U.S. or international), and undergraduate major. While the system is electronic, staff members are also allowed to make manual corrections to achieve balance.

Starting in 2009, with the advent of a new admissions director, IU's measures of diversity changed. The new system split students by application status (U.S. domestic, international, or U.S. underrepresented minorities), country of citizenship, gender, GMAT (generally defined as under 600, 600–690, and 700 and above), Keirsey Personality Type (Guardians, Artisans, Idealists, Rationals), and undergraduate major (defined as business studies, STEM disciplines, and all other subjects). In addition, special considerations were taken into account on rare occasions, usually requiring that two students in a relationship not be placed in the same cohort. Students are unable to switch cohorts/teams once the semester starts; however, students may be able to switch prior to the semester if the class time conflicts with childcare or a medical engagement.

While students are assigned to teams and cohort by balancing across characteristics, balancing does not negate our identification strategy because students are not assigned based on future employment goals. To understand this point it is necessary to recognize that by balancing teams, two randomly selected class peers are more likely (rather than less likely) to have similar characteristics than two cohort/team peers (Shue 2013). Assuming student characteristics predict employment outcomes, balanced assignment results in a bias against finding positive peer effects because these students are less likely to be interested in working at the same firm. As discussed later in the paper, we implement multiple robustness tests to confirm balancing is not driving our results.²⁷

4.1.3 Online business networking service. Without additional data sources, prior research on peer effects is often limited to short-term outcomes. To observe career outcomes over several years, we instead rely on LinkedIn

²⁶ For instance, according to Ahern, Duchin, and Shumway (2014), the University of Michigan MBA Program maximizes diversity within sections by equally weighting six dimensions: gender, ethnicity, citizenship, undergraduate institution, employer, and dual-degree status. Harvard Business School uses a similar randomization.

²⁷ First, we test whether placebo networks affect employment outcomes. To mimic the assignment process used by the MBA Admissions Office, we balance placebo teams based on the same characteristics used in the actual team/cohort assignments. As shown in Table 10, placebo teams have no effect on recruitment outcomes. Second, while employers cannot be assigned equally across teams, it is possible that connections to a particular industry correlate with variables that predict employment. Based on this concern, we evaluate whether connections to a given industry affect the likelihood of employment within that industry (absent the connected firm). According to the results in Table 11, we find no evidence that industry-level connections influence employment with other firms in the industry.

as a large online social network for both managers and firms.²⁸ We match each Kelley MBA graduate to his/her user profile on this online platform, which includes self-reported employment and education data. We then collect additional information about each employer, including the location and industry of the firm. All data are publicly available and are obtained through web searches and then parsed into a panel data set. We first identify all employment prior to joining the MBA. This means that internships completed prior to the MBA are included when we define connections, but internships completed between the first and second year of the MBA are excluded. Next, we identify employers within the first 5 years of MBA graduation.

4.1.4 Data matching. We manually match each MBA graduate to his or her online profile by first and last name, MBA degree, and year of graduation (whenever available in the profile). We drop any graduates without online profiles. To confirm that our matches are correct, we first require that each online profile refers to an MBA from the Kelley School of Business. In addition, we drop cases where the profiles list incorrect graduation years. Finally, for a portion of the sample (from graduating class 2004 to 2013), we have admissions data that includes undergraduate school. For this subsample, we also confirm that the undergraduate school from the admissions data matches the data from the online profile.

As MBA graduates heavily rely on LinkedIn for both employment and recruitment opportunities, we are able to match 95% of the 2,176 graduates in the sample. We include all 2,176 graduates when estimating the network effects in our empirical analysis. Including the graduates without a matching profile leads to more conservative estimates as we underestimate the number of prior connections as well as the number of people who joined top-tier (or mid-tier) firms after graduation.²⁹ Given that we are missing 5% of the individuals in our sample, one potential concern is that results might be biased if graduates with online profiles have different skills and/or interests compared to graduates without such profiles. We therefore summarize the match rate across demographics and career interests in Table A3 of the Internet Appendix. First, we estimate a 95% match rate for both men and women. Second, we have a slightly lower match rate of 92% among international graduates. Third, within American citizens, we match 95% of white graduates and 95% of minority graduates. Fourth, we find no evidence that predicted career interests (determined by intended major upon admission) have no impact on match

²⁸ Social networking data have a number of advantages over alternative sources. First, employer-employee-linked data from the U.S. Census Longitudinal Employer-Household Dynamics do not include employee names or educational histories (Jacobson, LaLonde, and Sullivan 1993; Graham et al. 2013); as a result, we cannot possibly identify our graduates at the individual level. Second, data on individual employees at public firms have been extensively used in the finance literature (Weisbach 1988; Jensen and Murphy 1990; Gompers, Ishii, and Metrick 2003); however, these data focus on top executives and exclude all information for employees at private firms.

²⁹ We note all empirical results continue to hold when we exclude the 5% of graduates without a LinkedIn profile.

Table 1
Summary statistics of MBA graduates

	N	Mean	Median	SD
Experience before MBA (years)	2,176	4.819	4.417	2.496
Mid- or top-tier firm before MBA	2,176	0.191	0.000	0.393
Mid- or top-tier firm after MBA	2,176	0.400	0.000	0.490
Prior employee	2,176	0.036	0.000	0.187
GMAT quantitative score	2,176	43.286	44.000	5.475
GMAT combined score	2,176	653.676	655.000	55.048
MBA classroom performance	2,007	85.266	85.868	4.721
Female	2,176	0.254	0.000	0.435
International	2,176	0.332	0.000	0.471
Racial minority	2,176	0.261	0.000	0.440
Team size	2,176	4.039	4.000	0.799
Cohort size	2,176	61.634	62.000	8.664
Graduation year	2,176	2,008.045	2,008.000	3.229

This table reports the summary statistics for MBA graduates graduating from the Indiana University MBA program in 2003–2013. The summary statistics for graduates are based on admissions information, employment histories, and cohort/team demographics. We note the smaller number of observations for the MBA Classroom Performance variable because of the data limitations described in the data section.

rate, suggesting our employment results would not be affected by the missing graduates.

4.2 Data summary of managers

4.2.1 Manager demographics. Our initial sample includes MBA students graduating between 2003 and 2013, comprising 2,176 graduates as presented in Table 1. Necessary to our analysis of social networks, the mean graduate has 4.8 years of work experience prior to the MBA. In addition, we estimate that 26% are female, 33% are international, and 26% are U.S. minorities. The mean graduate has a quantitative GMAT score of 43 out of 60 and a combined GMAT score of 654 out of 800. In addition, graduates receive an 85% in first-semester courses on average; we note we observe grades for only 2,007 graduates as we are missing data from the class of 2006. We also summarize the teams and cohorts in the sample. Students are randomly assigned to one of four cohorts during the early years of our sample period, and to one of three cohorts starting with the 2006 graduating class. Then, students are randomly assigned to teams within each cohort, resulting in a total of 40 to 50 teams per graduating class. According to the table, we estimate the average team comprises four students, while the average cohort comprises 62 students.

Last, we summarize the job titles of the MBA graduates in Figure 1. Analyzing the data collected from LinkedIn, we focus on six managerial titles: (1) Manager, (2) Director, (3) President or Vice President, (4) Executive, (5) Partner, and (6) Principal. Based on this definition, we estimate over 55% of graduates have a managerial job title within one year of graduation and this percentage increases to over 70% in year 5. While the majority of titles include the word “Manager,” we confirm a significant and rising fraction of higher-level

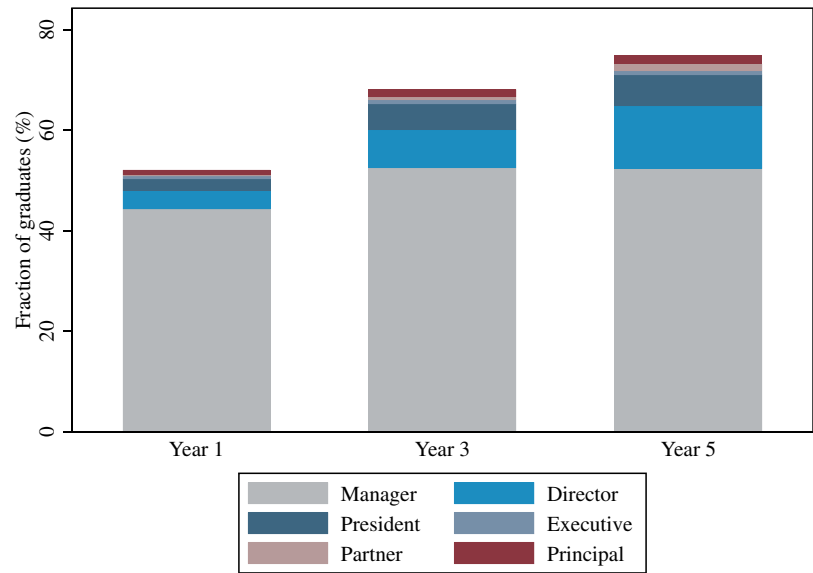


Figure 1
Composition of managerial job titles across MBA careers

This figure plots the breakdown of professional titles among Indiana University MBA graduates. We plot titles at the time of MBA graduation, 3 years after graduation, and 5 years following graduation. The plots include the titles Manager, Director, President, Executive, Partner, and Principal.

titles over the 5-year period: by year five, 14% of graduates hold Director-level jobs, and 7% of graduates are either “Presidents” or “Vice Presidents.”

4.2.2 Measures of predicted job performance. We measure predicted job performance in our setting as cognitive ability, also known as general mental ability. The personnel psychology literature has long recognized the unique role of cognitive ability in predicting job performance when compared to all other measures (Hunter 1986; Ree and Earles 1992). For instance, according to Schmidt and Hunter (1998), “Today, the validity of different personnel measures can be determined with the aid of 85 years of research. The most well known conclusion from this research is that for hiring employees without previous experience in the job the most valid predictor of future performance and learning is general mental ability.”³⁰ More specifically for our purposes, cognitive ability is a common measure of ex ante predicted job performance in past social networks research including Hensvik and Skans (2016).

We measure cognitive ability based on scores from the quantitative section of the Graduate Management Admissions Test, or GMAT. As the GMAT is

³⁰ In addition, Schmidt and Hunter (2004) argues “GMA [general mental ability] predicts both occupational level attained and performance within one’s chosen occupation and does so better than any other ability, trait, or disposition and better than job experience.”

the standard entrance exam required by all MBAs across the United States, all graduates in our sample have a GMAT score. Given that MBA rankings depend both directly and indirectly on the average exam scores of the entering MBA students, MBA programs have significant incentives to base admissions decisions on entrance exam scores.³¹ Potential MBA applicants therefore have strong wage incentives to earn a high score and gain acceptance to a highly ranked MBA program. The standard GMAT score, which has a range of 200–800, is based on a verbal section and quantitative section.³² As nonnative speakers likely face a disadvantage in the writing and verbal sections, we focus the majority of our results on the quantitative score, which has a score range of 0–60. However, we illustrate all results hold when we instead focus on the combined GMAT score.

We also consider an alternative measure: MBA classroom performance based on grades in first-semester courses. In the first semester, all MBA students are enrolled in the Core Curriculum. These courses cover eight topics taught by eight different faculty members: Critical Thinking, Economics, Finance, Accounting, Marketing, Operations, Quantitative Analysis, and Strategic Management. We believe there are two benefits to focusing on academic grades in the MBA program. First, a recent literature has confirmed the value of general managerial skills over more firm- or industry-specific knowledge (Frydman 2005; Murphy and Zabochnik 2007; Custódio, Ferreira, and Matos 2013). Therefore, an MBA education is likely valuable across the range of firms and industries in our sample. Second, shareholders appear to value educational credentials in managers as stocks react more positively when a newly appointed manager has higher educational credentials, and these managers receive higher compensation (Bhagat, Bolton, and Subramanian 2010; Falato, Li, and Milbourn 2015). This preference is unsurprising given that managers with MBAs have superior performance to peers without an MBA (Chevalier and Ellison 1997; King, Srivastav, and Williams 2016).

5. Methodology

First, we develop an empirical framework that allows us to identify the impact of social networks on managerial hiring outcomes in the sample; second, we outline our measures of cognitive ability; and third, we discuss the set of firms included in our analysis.

5.1 Empirical specification

To identify whether social networks have a causal effect on managerial hiring outcomes, we must address the endogeneity associated with the formation of

³¹ For instance, in the 2018 MBA rankings, over 16% of the rankings are based on the mean entrance exam scores.

³² Technically, the GMAT has four sections: an analytical writing assessment, an integrated reasoning section, a quantitative section, and a verbal section. However, the analytical writing and integrated reasoning section are graded separately and are not used in the standard GMAT score.

such networks. In our analysis, a manager has a connection to a firm if she knows a former employee of that firm. However, manager connections are potentially driven by particular skill sets, current and past decisions, and networking efforts (Manski 1993; Hellerstein, Kutzbach, and Neumark 2015). To estimate plausible causal effects of former employees' networks, we rely on the forced assignment into teams and cohorts introduced by the MBA office at Indiana University while current managers were enrolled in the MBA program. Past research has developed similar strategies to identify peer effects among cohort member in MBA programs (Lerner and Malmendier 2013; Ahern, Duchin, and Shumway 2014; Shue 2013), yet we are the first to analyze a more closely defined definition of peers based on teams of roughly four students.³³ Students in the same cohort take the core MBA classes together, while students in the same team are assigned to work together on course projects and a large case study at the end of the semester. Importantly, as the forced assignment does not depend on graduates' pre-MBA experience, the set of connections is exogenous to the job applicant.

For the purposes of our regression analysis, we compile our data set into a cross-section structure where we observe the employment of each manager at the annual level for the first 5 years following graduation. In addition, we also identify all places of employment for each individual prior to entering the Indiana University MBA program. With these data, we can evaluate employment choices under a general model of choice. Economics has long recognized the value of these models to evaluate migration decisions; more recently, finance papers have implemented these methodologies to understand the effects of venture capital networks (Hellmann, Lindsey, and Puri 2008; Lindsey 2008), though these econometric frameworks remain rare in finance. By extending these econometric methods to understand the role of managerial networks, we provide a framework for future research in managerial labor markets.

We assume individual i chooses employment among J alternative firms. Let U_{ij} represent the utility of individual i matching with employer j . Utility U_{ij} is treated as an independent random variable with a systematic component ζ_{ij} and a random component ε_{ij} such that: $U_{ij} = \zeta_{ij} + \varepsilon_{ij}$. In the standard logistic model, the expected utilities ζ_{ij} are modeled simply as characteristics of the individuals $\zeta_{ij} = x_i \eta_j$. We instead estimate an alternative-specific conditional logistic regression as in McFadden (1973), which models the expected utilities in terms of the (a) individual-specific attributes and (b) alternative-specific characteristics. As such, $\zeta_{ij} = x_i \eta_j + z_{ij} \beta$, where z_{ij} represents characteristics varying across choices and individuals.

³³ More specifically, Shue (2013) studies how CEO/CFO peers affect firm policies, Ahern, Duchin, and Shumway (2014) focuses instead on measures of altruism and trust. Our paper is more similar to that of Lerner and Malmendier (2013), who consider a separate employment outcome, namely, entrepreneurship at graduation. A related line of research relies on random assignment of dormmates and roommates to examine their influence on short-term test scores (Sacerdote 2001; Zimmerman 2003; Lyle 2009; Carrell, Fullerton, and West 2009).

For our purposes, we are interested in understanding whether graduate i is more likely to match with employer j when connected to employer j . We identify a connection when any team members (or cohort members, in robustness tests) of individual i worked at firm j prior to entering the MBA program. Prior employment includes all employment experiences, including summer internships.³⁴ The alternative-specific conditional logistic model is therefore necessary as connections vary across both employers and individuals. Following McFadden (1973), we then model the probability that graduate i is employed at each firm j as follows:

$$\Pr(Y_i = j) = \frac{e^{\zeta_{ij}}}{1 + \sum_{m=1}^{J-1} e^{\zeta_{im}}} = \frac{e^{\eta_j \cdot \mathbf{X}_i + \beta \cdot \mathbf{Z}_{j,i}}}{1 + \sum_{m=1}^{J-1} e^{\eta_m \cdot \mathbf{X}_i + \beta \cdot \mathbf{Z}_{m,i}}}, \text{ for all } j = 1, \dots, J-1$$

$$\Pr(Y_i = J) = \frac{1}{1 + \sum_{m=1}^{J-1} e^{\eta_m \cdot \mathbf{X}_i + \beta \cdot \mathbf{Z}_{m,i}}}, \text{ for all } j = J, \quad (1)$$

where the last alternative (J) is defined as the base alternative. We discuss below how we define the base alternative in our setting and interpret the coefficient of interest β . The alternative- and individual-specific characteristics define the probability that graduate i is employed at firm j . These characteristics are modeled as:

$$\eta_j \cdot \mathbf{X}_i + \beta \cdot \mathbf{Z}_{j,i} = \underbrace{\alpha + \beta \times \text{Conn}_{ij} + \text{Firm} \times \text{Year FE}}_{\beta \cdot \mathbf{Z}_{j,i}} + \underbrace{\eta_j \times \text{Manager-controls}_i}_{\eta_j \cdot \mathbf{X}_i}, \quad (2)$$

where the independent variable of interest is Conn_{ij} , a binary variable denoting whether any team (or cohort) member of individual i held prior employment with firm j . For individuals who are former employees, we subtract Conn_{ij} by one.³⁵ To test Hypothesis 1, we determine whether $\beta > 0$, which implies that a connection through former employees support the firm's recruitment process.

We include two sets of controls in our analysis. First, the likelihood of entering a given firm differs significantly over the cross-section of employers: certain firms regularly hire five or more managers each year, while other firms rarely hire even one manager. In addition, a graduate's set of job offers is dependent on the particular requirements of that employer; some firms make offers to a wide set of applicants, while other firms only hire the most capable graduates. To overcome these issues, we include firm-year fixed effects in our specifications. Second, social connections are only random after

³⁴ In robustness tests, we vary this definition based on the length of tenure of the former employee.

³⁵ For teams this effectively reduces the Conn_{ij} to zero since the instance of two individuals from the same firm in the same team is rare.

controlling for the variables used in the assignment process. As students are assigned to cohorts and teams based on specific characteristics, we control for demographic differences across gender, race, nationality, and GMAT (split into two quantiles).

We estimate our discrete choice model using a maximum likelihood function, which is maximized using a combination of the modified Newton-Raphson and the Davidon-Fletcher-Powell algorithms.³⁶ To address correlations between error terms across managers, we impose the variance-covariance matrix to be the observed information matrix, or OIM. The OIM is the negative of the second derivative (the Hessian matrix) of the log-likelihood (the logarithm of the likelihood function), estimated at the point where maximum likelihood function is maximized. In robustness tests reported in Table A10, we consider alternative clustering techniques, namely, robust errors and clustering at the team, cohort, and year levels. We confirm clustering under OIM results in the most conservative standard errors.

5.2 Sample of employers

As there are a total of 974 firms hiring the 2,176 graduates, the estimation of a discrete choice model is not computationally feasible and results in a misspecified model (White 1982). To solve this issue, we start by focusing on the subset of potential employers with multiple connections to prospective managers: in other words, firms that employed at least two individuals from the classes of 2003–2013 prior to joining the Kelley MBA. This choice allows us to focus on 121 firms that recruit nearly half of IU MBA graduates. The excluded firms are typically small and local firms that only occasionally hire MBA graduates. We collapse the firms that are not modeled explicitly into an alternative employer denoted other firm.”³⁷

Our tests with the sample of 121 firms still produce an unstable maximum likelihood function unable to converge to a maximum solution. This instability of the likelihood functions stems from the large number of coefficients (605 in total) that need to be estimated as our model requires estimating one coefficient per alternative for each individual-specific control variable. Including all control variables used in the random assignment process therefore leads to a misspecified model because of the high ratio of parameters to the sample size. Instead, we find the inclusion of roughly 40 alternatives in the estimation provides a stable optimization process of the maximum likelihood function. In line with this approach, several recent studies in the migration literature

³⁶ In most specifications, we use 20 interactions of the Newton-Raphson algorithm followed by 30 interactions of the Davidon-Fletcher-Powell algorithm.

³⁷ We are required to include the other firm” as discrete choice models require that the decision-maker necessarily chooses one of the alternatives (Train 2009). However, in unreported results, we exclude the other firm” in a linear regression framework. As we find minimal differences in our estimates, we argue the inclusion of the other firm” is not driving the results.

use similar sample sizes to study state-to-state migration, which include 50 alternative choices (Groen 2004; Kennan and Walker 2011). As such, we divide our sample of 121 firms into three tiers and try to take advantage of the split to provide further evidence for our main test: Hypothesis 3.

Hypothesis 3 relies on the implicit prediction that the effect of social networks should disproportionately affect individuals with high predicted job performance as measured by cognitive ability. Our ranking methodology therefore sorts firms based the importance of cognitive ability in firm hiring decisions. We propose using companies that disproportionately hire managers graduating from the best U.S. MBA programs, which have the highest admissions standards. To identify these firms, we introduce employment data for MBA graduates graduating from the three top-ranked MBA programs during the period 2001–2013: Harvard University, University of Chicago, and Stanford University.³⁸ We argue firms hiring from top-ranked MBA programs require managers with high cognitive ability: according to recent admissions statistics, the mean score in the Harvard University, University of Chicago, and Stanford University MBA Class of 2019 was 730, 726, and 734, respectively. For comparison, this corresponds to the 90th percentile of the 2003–2013 Indiana University MBA programs.³⁹

The sample of 121 firms is then divided into three tiers based on the number of managers hired from each of the three top-ranked MBA programs. *Top-tier* firms hired at least four or more graduates from each of these programs, *mid-tier* firms hired 2 to 3 graduates from each program, and *bottom-tier* firms hired at most one graduate from each program.⁴⁰ Bottom-tier firms not only rarely hire from top MBA programs but also have a low hiring rate of IU graduates, a fact that limits the number of hypotheses that we can test with this sample. Therefore, in the paper we report all tests for top- and mid-tier firms, and in the Internet Appendix we report the results that are econometric feasible for bottom-tier firms—mainly those that do not require further subsample tests. The set of top-tier firms includes 43 firms, the set of mid-tier firms includes 38 firms, while the bottom-tier firms contain 40 firms.

Table 1 reports the connections that IU MBAs have to top- and mid-tier firms. We estimate 19% of the sample previously worked at a top- or mid-tier firm prior to joining the MBA. In addition, 40% of graduates join one of these firms after graduation. In total, each graduate in the sample is connected to be between 0 and 3 of these employers through a team member and between 1 and

³⁸ This three programs include the highest-ranked U.S. MBA programs across the East Coast, Midwest, and West Coast. For example, in the 2017 *U.S. News* MBA rankings, the Harvard MBA is ranked first, the Chicago MBA is second, and the Stanford MBA is fourth. In the 2017 *Bloomberg* MBA rankings, the Harvard MBA is ranked first, the Stanford MBA is second, and the Chicago MBA is fourth. Rankings in previous years are similar.

³⁹ In unreported results, we confirm firms that most frequently hire from top-ranking MBA programs also hire the Indiana University MBA graduates with the highest quantitative GMAT scores, combined GMAT scores, and classroom performance. These results suggest the firms are especially likely to measure talent based on cognitive ability.

⁴⁰ Our results do not depend on this breakdown. For instance, tiers are similar when we split the sample based on total number of hires across all firms or number of hires in a given year.

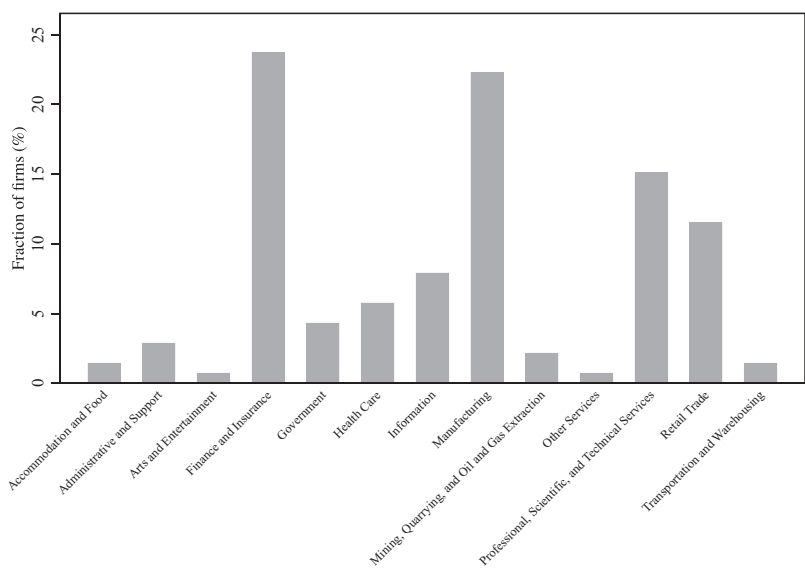


Figure 2
Industry composition of top and mid-tier firms

This figure plots the industry classification for all top and mid-tier firms. Industry classification is based on two-digit NAICS codes. Top-tier firms are all firms that employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and employed at least four MBA graduates from each of the three top-ranked MBA programs between 2003–2013. Mid-tier firms are all firms that (employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and employed 2 to 3 MBA graduates from each of the three top-ranked MBA programs between 2003 and 2013).

21 of these firms through a cohort member. We match each of these firms to a two-digit NAICS industry classification and present the industry breakdown in Figure 2. The most common industries are (a) finance and insurance (19% of firms included in the sample); (b) retail trade (14%); and (c) professional and technical services (13%), information (13%), and educational services (13%). Our sample includes firms across 13 industries. Figure 3 plots the distribution of employment size for each tier of firms. In line with the arguments above, we find similar employment levels among top- and mid-tier firms with a median of roughly 75,000 workers; in contrast, bottom-tier firms are smaller with a median size of roughly 10,000 employees.

For further information, we list the full set of firms and characteristics in Table A1 and Table A2 of the Internet Appendix. The five firms hiring the largest number of Indiana University graduates include EY (72 graduates), Proctor & Gamble (71), Cummins, Inc. (70), Eli Lilly (61), and GE (60). The firms employing the greatest number of graduates prior to the MBA include Accenture (41 graduates), PwC (38), Deloitte (30), Infosys (28), EY (26), and Eli Lilly (22). The median firm previously employed four IU graduates, though the mean is higher at 6.6 graduates.

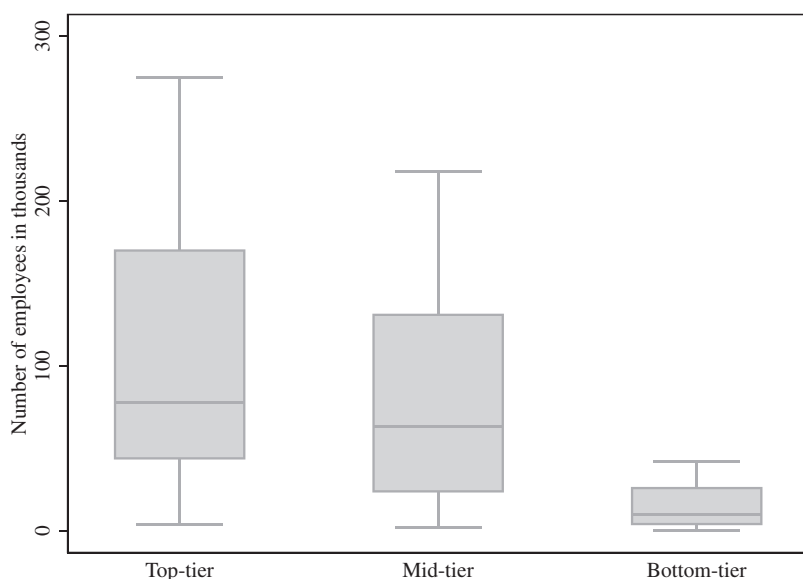


Figure 3
Distribution of firm size by tier

This figure plots the size distribution for all top-, mid-, and bottom-tier firms. Employment figures are estimated using LinkedIn. Top-tier firms are all firms that employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and employed at least four MBA graduates from each of the three top-ranked MBA programs between 2003 and 2013. Mid-tier firms are all firms that employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and employed 2 to 3 MBA graduates from each of the three top-ranked MBA programs between 2003 and 2013. Bottom-tier firms are all firms that employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and employed 0 to 1 MBA graduates from each of the three top-ranked MBA programs between 2003 and 2013.

6. Results

First, we evaluate the impact of managerial social networks on firms. Second, we test whether employment outcomes are influenced more by weak or strong ties. Third, we investigate which firms most benefit from social networks. Fourth, we examine the heterogeneous effects of social networks among low- and high-ability managers. Fifth, we discuss survey evidence outlining the mechanisms relating social networks and managerial recruitment outcomes. Sixth, we consider econometric concerns with our analysis. Lastly, we present evidence rejecting alternative explanations of the results.

Throughout this section, we report our results of the discrete model as odds ratios (the exponent of the coefficient reported in our tables) according to Equation (2). For the baseline results, we report the implied probabilities as defined in Equation (1). We do not report the implied probabilities for all results as it would require estimating the second term of Equation (1) for specific values of the control variables and fixed effects.

6.1 Do social networks affect the recruiting outcomes of managers?

6.1.1 Team analysis. Table 2 evaluates the role of social networks on recruitment under a discrete choice model. Panels A and B include all employment choices within top-tier and mid-tier firms, respectively. Columns 1 through 3 focus on employment 1 year after graduation; later columns focus on employment 5 years after graduation. All columns include a firm-year fixed effect, given firm hiring demand varies across years and randomization is only possible within a graduating class. In addition, columns 2, 3, 5, and 6 include controls for characteristics used in the random assignment of students into teams, namely, gender, race, nationality, and GMAT score. For the purposes of the regression, nationality identifies domestic versus international students; race is defined only for domestic students, and identifies whites versus non-whites; and GMAT fixed effect identifies students above and below the median GMAT score in the sample. Estimates are reported as an odds ratio, as defined in Equation 2, which is the effect that one connection has on the likelihood of getting a job at firm j in top-tier (mid-tier) relative to the effect that a connection has on the likelihood of getting a job at a firm outside the top-tier (mid-tier), the base alternative. All columns include 2,176 observations as we include the 5% of graduates with a missing LinkedIn profile. We note that this choice leads to conservative estimates since we underestimate the number of connections and number of graduates that join top- and mid-tier firms. Nonetheless, all regression estimates are similar when we exclude these graduates.

According to column 1 of panel A, we estimate that the effect of a connection on the likelihood of joining a top-tier firm is 2.22 ($=e^{0.796}$) times larger than the counterpart effect for firms outside the top-tier. This odds ratio translates into a large economic effect given we estimate that a connection to firms outside the top-tier increases the likelihood of employment for an average manager by 1.6 percentage points—this value corresponds to the second term on the right-hand side of Equation (1).⁴¹ A connection to a top-tier firm would then increase the likelihood of employment by 3.5 ($=1.6 \times 2.22$) percentage points. After including demographic controls in column 2, the effect remains unchanged. Columns 4 and 5 evaluate employment at 5 years after graduation. As we estimate a similar odds ratio of 1.91 ($=e^{0.689}$), we find connections continue to affect the allocation of managers even 5 years after graduation. The results suggest that social networks have a persistent effect on the distribution of managers across firms.⁴²

We find similar, but slightly larger effects, for mid-tier firms. As reported in columns 1 and 2 of panel B, the odds ratio of the effect of a connection on the

⁴¹ We estimate this probability for the average values of the control variables.

⁴² In unreported results, we instead group firms based on (a) industry classification and (b) employment size and then reestimate the discrete choice models. As we continue to find evidence that connections increase the likelihood a hiring a given manager, these results help confirm our findings do not depend on exactly how we group firms in the discrete-choice framework.

Table 2
Network effects

A. Network effects across top-tier employers

	1 year			5 years		
	(1)	(2)	(3)	(1)	(2)	(3)
Connection	0.796*** (3.48)	0.790*** (3.40)		0.650** (2.51)	0.689*** (2.65)	
Long-tenure connection			0.989*** (3.05)			0.868** (2.44)
Short-tenure connection			0.588* (1.75)			0.473 (1.21)
Gender FE	No	Yes	Yes	No	Yes	Yes
Race FE	No	Yes	Yes	No	Yes	Yes
Nationality FE	No	Yes	Yes	No	Yes	Yes
GMAT FE	No	Yes	Yes	No	Yes	Yes
Firm × year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2,176	2,176	2,176	2,176	2,176	2,176

B. Network effects across mid-tier employers

	1 year			5 years		
	(1)	(2)	(3)	(1)	(2)	(3)
Connection	0.997*** (2.92)	1.094*** (2.94)		0.785* (1.93)	0.761* (1.85)	
Long-tenure connection			1.139** (2.41)			0.701 (1.40)
Short-tenure connection			0.962** (1.84)			0.875 (1.30)
Gender FE	No	Yes	Yes	No	Yes	Yes
Race FE	No	Yes	Yes	No	Yes	Yes
Nationality FE	No	Yes	Yes	No	Yes	Yes
GMAT FE	No	Yes	Yes	No	Yes	Yes
Firm × year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2,176	2,176	2,176	2,176	2,176	2,176

This table presents the team-level network results under a discrete choice model. The dependent variable denotes the firm that hired the given manager. Panel A considers employment options across top-tier firms, while panel B considers employment options across mid-tier firms. Top-tier firms are all firms that employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and employed at least four MBA graduates from each of the three top-ranked MBA programs between 2003 and 2013. Mid-tier firms are all firms that employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and employed 2 and 3 MBA graduates from each of the three top-ranked MBA programs between 2003 and 2013. Columns 1–3 measure employment 1 year after graduation; columns 4–6 measure employment 5 years after graduation. Columns 1–2 and 4–5 define the network at the team level and define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular team. Columns 3 and 6 distinguish between whether the prior employee has a short or long-tenure relationship with the prior employer. A strong relationship is defined as team connections to a former employee who worked for 3 or more years in the elite firm prior to the MBA. Controls include fixed effects for graduation year × firm, gender, nationality, GMAT score (split at the median), and race. Standard errors are computed under the observed information matrix. * $p < .1$; ** $p < .05$; *** $p < .01$.

likelihood of join a mid-tier firm is 2.99 ($=e^{1.094}$); the coefficients are similar for employment 5 years after graduation. Table A7 in the Internet Appendix reports the same estimation for bottom-tier firms. There we find that the odds ratio of the effect of a connection on the likelihood of join a bottom-tier firm

4.54. In line with Hypothesis 1, we find continued evidence that social networks influence recruitment across a wide range of employers.⁴³

6.1.2 Relationship to firm. The results above confirm social networks have a positive influence on the likelihood of hiring a given manager. We argue these results are driven by social networks facilitating the flow of information between firms and potential applicants. In support of this argument, we next split former employees based on their tenure with the firm. Assuming former employees provide information to firms, employees that remained longer with the firm have more contacts with the employer and are more trusted to provide accurate referrals. Assuming former employees provide information to applicants, workers that remained longer with the firms will have more accurate information about the firm. Therefore, regardless of the direction of information, we should find stronger network effects about employees with longer tenures.

We present these findings in columns 3 and 6 in panel A of Table 2. We first split the sample based on the number of years for which a connection was employed at the hiring firm. In our sample, the median tenure with an employer prior to the MBA is 3 years. We define the prior employee-employer relationship as long-term if the manager was previously employed at the firm of interest for at least 3 years, and as short-term if the manager was employed there for less than 3 years. Estimating the baseline model for top-tier firms separately for each subsample, we find that the odds ratio of having a long-term connection is 2.69. In contrast, the odds ratio of having a short-term connection is smaller at 1.80.⁴⁴ The results offer evidence that the positive influence of social networks on recruiting outcomes is likely a consequence of former employees holding valuable information.

6.2 Do social networks affect the recruitment of managers for *less-known* firms?

6.2.1 Frequent and infrequent recruiters. According to the preceding results, social networks increase the likelihood a firm hires a given manager; we next determine *which* firms are more reliant on these networks. Assuming social networks support the flow of information to potential applicants, Hypothesis 2a suggests networks are especially relevant for firms with less opportunity to provide and receive information through alternative channels. By splitting firms into high-, mid-, and low-tier firms based on the number of managers hired from

⁴³ These findings are also consistent with Hypothesis 2a as we find that social connections are more valuable for firms that infrequently hire managers hired from top MBA programs. These firms are likely less known and therefore depend on social networks to provide and receive information in the recruiting process.

⁴⁴ In unreported results, we estimate whether the effects of long- and short-tenure connections are statistically different under a linear probability model. Estimating employment across both top- and mid-tier firms, the difference is statistically significant at the 5% level.

Table 3
Network effects across more and less-known firms

	Hiring frequency		HQ distance from IU	
	High	Low	Local	Nonlocal
Connection	0.780*** (3.47)	1.413** (2.38)	0.698** (2.30)	0.858*** (3.04)
Gender FE	Yes	Yes	Yes	Yes
Race FE	Yes	Yes	Yes	Yes
Nationality FE	Yes	Yes	Yes	Yes
GMAT FE	Yes	Yes	Yes	Yes
Firm × year FE	Yes	Yes	Yes	Yes
N	2,029	1,554	1,800	1,784

This table presents team-level network results under a discrete choice model across firm characteristics. The dependent variable denotes the firm that hired the given manager within 1 year of graduation. We define the network at the team level and define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular team. Column 1 reports the network regression results when the employment choices are firms that (a) employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and (b) hired at least 10 MBA graduates in the classes of 2003–2013 within 1 year following graduation. Column 2 reports the network regression results when the employment choices are firms that (a) employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and (b) have headquarters within 500 miles from Bloomington, Indiana. Column 3 reports the network regression results when the employment choices are firms that (a) employed multiple IU MBA graduates in the classes of 2003–2013 prior to joining the MBA and (b) have headquarters located more than 500 miles away from Bloomington, Indiana. Controls include fixed effects for graduation year × firm, gender, nationality, GMAT score (split at the median), and race. Standard errors are computed under the observed information matrix. * $p < .1$; ** $p < .05$; *** $p < .01$.

top-ranked MBA programs, we have already demonstrated networks effects are greatest for low-tier firms in infrequently hire from these programs and smallest for high-tier firms that frequently hire from these programs. Assuming hiring from top-ranked programs is a proxy for the information graduates have about these employer, our results already provide evidence in support of Hypothesis 2a. Next, we develop two additional methods to identify less-known employers.

First, we identify employers that infrequently recruit Indiana University MBA graduates. We argue potential applicants to these firms will have limited access to information through recent alumni and on-campus recruiting events. Under our firm heterogeneity tests, we group top-, mid-, and bottom-tier firms together to allow for a wide range of variation. We then split all firms into two groups: frequent recruiters are those hiring at least 10 MBA graduates from the classes of 2003–2013. According to the first and second columns of Table 3, we find that among frequent recruiters the odds ratio of team member connection is 2.18. In contrast, for infrequent employers the odds ratio of a team member connection is 4.11. These results confirm Hypothesis 2a.

6.2.2 Local and nonlocal firms. Second, we analyze geographic distance between the Indiana University campus in Bloomington, Indiana, and the employer’s headquarters. We then classify local firms as those in which the headquarters are located less than 500 miles from Bloomington, Indiana.

According to the third and fourth columns of Table 3, we find for local firms the odds ratio of team member connection is 2.01. In contrast, for nonlocal employers the odds ratio of a team member connection is 2.36. The results confirm social networks disproportionately affect nonlocal firms, further reinforcing Hypothesis 2a.⁴⁵

6.3 Do strong social networks affect the recruiting outcomes of managers?

6.3.1 Cohort analysis. Thus far, our analysis has identified social networks at the team level: teams comprise small groups of four members who work on course assignments together in addition to a case competition at the end of the first semester. However, our data also include information on cohorts, groups of roughly 60 students who take the first semester coursework together. Next, we evaluate the differential effect of cohort and team connections. Under Hypothesis 2b outlined above, we argue that strong connections should have a larger impact on employment outcomes, while weaker connections have smaller employment effects because of less common interactions.

We present our findings in panel A of Table 4. Columns 1 through 4 consider employment choices across top-tier firms, while columns 5 through 8 consider employment choices across mid-tier firms. Columns 1, 2, 5, and 6 consider employment within a year following graduation, while the remaining columns consider employment 5 years after graduation. As illustrated in columns 1 and 2, we estimate that the odds ratio of a cohort-level connection is 1.40 for employment in a top-tier firm within 1 year of graduation. The estimates are quantitatively similar 5 years out. We also find similar estimates when considering mid-tier firms: we estimate the odds ratio to be 1.71 for employment 1 year after graduation. For comparison, in Table 2 we estimated a team-level connection odds ratio for top-tier firms to be 2.99.⁴⁶ Therefore, consistent with Hypothesis 2a, while cohort members affect managerial recruitment outcomes, the effect of a team-level connection is larger than a cohort-level one.

6.3.2 Aggregate network effects. We also use an alternative economic specification that more closely follows past specifications in the corporate governance literature. As illustrated by Kramarz and Thesmar (2013), social network models at the individual level can be readily aggregated to the level of the network. In our particular setting, we estimate the proportion that a given group joins the connected employer. A primary benefit of this aggregated

⁴⁵ In unreported results, we estimate whether the effect of a connection with a well-known firm is statistically different from a connection with a less-known firm under a linear probability model. Across both top- and mid-tier firms, the difference is statistically significant at the 5% level.

⁴⁶ In unreported results, we estimate whether the effect of a team-level connection is statistically different from a cohort-level connection under a linear probability model. We estimate the difference is statistically different at the 1% level.

Table 4
Cohort-level network effects

A. Cohort-level network effects

	Top-tier				Mid-tier			
	1 year		5 years		1 year		5 years	
Connection	0.334*** (2.74)	0.336*** (2.73)	0.280** (2.08)	0.304** (2.24)	0.439** (2.35)	0.539*** (2.59)	0.509** (2.36)	0.622*** (2.86)
Gender FE	No	Yes	No	Yes	No	Yes	No	Yes
Race FE	No	Yes	No	Yes	No	Yes	No	Yes
Nationality FE	No	Yes	No	Yes	No	Yes	No	Yes
GMAT FE	No	Yes	No	Yes	No	Yes	No	Yes
Firm × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	2,176	2,176	2,176	2,176	2,176	2,176	2,176	2,176

B. Aggregate network effects

	All firms		Top and mid-tier		Top-tier		Mid-tier	
	Team	Cohort	Team	Cohort	Team	Cohort	Team	Cohort
Team connection	0.020*** (5.35)		0.019*** (3.80)		0.018** (2.88)		0.028*** (3.03)	
Cohort-connection		0.002*** (3.55)		0.002** (2.51)		0.002 (1.61)		0.002* (1.98)
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm × year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	69,052	4,392	46,412	2,952	24,904	1,584	22,074	1,404

This table presents the cohort-level network results. Panel A analyzes network effects under a discrete-choice framework. The dependent variable denotes the firm that hired the given manager. Columns 1–4 report the network regression results when the employment choices are top-tier firms, while columns 5–8 report the network regression results when the employment choices are mid-tier firms. Columns 1–2 and 5–6 measure employment 1 year after graduation; columns 3–4 and 7–8 measure employment 5 years after graduation. We define network at the cohort-level and define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular cohort. Controls include fixed effects for graduation year × firm, gender, nationality, GMAT score (split at the median), and race. Standard errors are computed under the observed information matrix. Panel B analyzes the aggregate network effects under a linear probability framework. Columns 1, 3, 5, and 7 aggregate the data at the team level, and the dependent variable denotes a member of the team is hired by the given firm within 1 year of graduation. Team connection is a binary variable that takes a value of one when the firm previously employed a student in that particular team. Columns 2, 4, 6, and 8 aggregate the data at the cohort-level and the dependent variable denotes a member of the cohort is hired by the given firm within 1 year of graduation. Cohort-connection is a binary variable that takes a value of one when the firm previously employed a student in that particular cohort. Columns 1 and 2 report the network regression results when the employment choices are across top-, mid-, and bottom-tier firms; columns 3 and 4 report the network regression results when the employment choices are top- and mid-tier firms; columns 5 and 6 report the network regression results when the employment choices are top-tier firms; and columns 7 and 8 report the network regression results when the employment choices are mid-tier firms. Controls include fixed effects for graduation year × firm; columns 1, 3, and 5 also include cohort fixed effects. Standard errors are clustered separately at both the year level and the firm level. * $p < .1$; ** $p < .05$; *** $p < .01$.

approach is that we reduce the number of null employment observations. By completing this analysis separately at both the team level and cohort level, we can estimate the effect of both strong ties and weak ties on recruitment. We estimate all network effects under a linear regression, and cluster the errors separately at both the firm and year level.

We present our results in panel B of Table 4. In columns 1, 3, 5, and 7, we aggregate networks at the team level, while in columns 2, 4, 6, and 8, we

aggregate networks to the level of the cohort. By moving from the discrete-choice framework, we are able to expand the number of potential employers. Therefore, we consider employment across top-, mid-, and bottom-tier firms in columns 1 and 2, and employment across both top- and mid-tier firms in columns 3 and 4. In addition, we consider exclusively employment in top-tier firms in columns 5 and 6 and mid-tier firms in columns 7 and 8. According to column 1, we estimate a team previously connected to a given employer is 1.9 percentage points more likely to have at least one other team member join a given top- or mid-tier employer relative to an unconnected team. According to the second column, we estimate a connected cohort is 0.2 percentage points more likely to have at least one cohort member join the employer than an unconnected employer. Both results are statistically significant at the 1% level. The results are similar when focusing on just top- and mid-tier firms separately. The results further verify the role of strong ties relative to weak ties under an alternative empirical specification.

A related, but separate question is whether the combined role of all cohort members influences future employment relative to the combined role of all team members. The argument is straightforward: while the effect of a teammate is larger, there are fewer teammates and more cohort members. We estimate the mean team has 4 members (and therefore each student has 3 teammates), while the mean cohort has roughly 60 members. Multiplying the individual cohort member effect of 0.2 percentage points by twenty results in a cumulative cohort effect of 4.0 percentage points; this is larger, though similar in magnitude, to the team member effect estimated at 1.9 percentage points. Our results again confirm the findings of Gee, Jones, and Burke (2017) that weak ties have a greater combined impact on employment relative to the combination of strong ties.

6.3.3 Within-team variation. For additional evidence that an individual strong tie has a greater impact than an individual weak tie, we exploit within-team variation. We consider especially strong ties, team members that also live near each other while attending Indiana University. We first note these groups are not random, but endogenous as managers choosing to live in the same area may also share other similarities, including wealth and income, or preferences for particular amenities. In this instance, we are not identifying a network effect, but rather a “similarity” effect. Despite this potential weakness, we believe there remains value in developing a measure of especially strong social ties. For every teammate pair, we determine whether the pair lived within 1.5 miles of each other in the first year of the MBA (defined as geographically close team members), or otherwise (defined as distant team members). We use 1.5 miles to distinguish distant and close peers as it is near the median distance between any two MBA graduates. Assuming students living near each other are more likely to regularly interact outside of class, we argue these students

Table 5
Team network effects under strong and weak ties

	Top-tier firms				Mid-tier firms			
	Peer residence		Peer gender		Peer residence		Peer gender	
	Close	Distant	Same	Different	Close	Distant	Same	Different
Connection	1.267*** (3.09)	0.508 (1.62)	1.185*** (4.28)	-0.801 (-1.03)	1.883*** (3.12)	0.485 (0.96)	1.466*** (3.71)	0.959 (-0.41)
Gender FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Race FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nationality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
GMAT FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm × year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	2,176	2,176	2,176	2,176	2,176	2,176	2,176	2,176

This table presents both strong and weak team-level network results under a discrete choice model. The dependent variable denotes the firm that hired the given manager within 1 year of graduation. Columns 1–4 report the network regression results when the employment choices are top-tier firms, while columns 5–8 report the network regression results when the employment choices are mid-tier firms. We define the network at the within-team level. In columns 1 and 5, we define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular team and the students live within 1.5 miles of each other at the time of entering the Indiana University MBA program. In columns 2 and 6, we define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular team and the students live more than 1.5 miles of each other at the time of entering the Indiana University MBA program. In columns 3 and 7, we define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular team and the students share the same gender. In columns 4 and 8, we define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular team and the students are different genders. Controls include fixed effects for graduation year $\times$ firm, gender, nationality, GMAT score (split at the median), and race. Standard errors are computed under the observed information matrix. * $p < .1$; ** $p < .05$; *** $p < .01$.

should also have a greater influence on employment outcomes. We test this argument in Table 5.

According to column 1, we estimate having a connection through a close team member leads to an odds ratio of 3.55. In comparison, have a distant team member implies an odds ratio of 1.66 according to column 2. The results are even larger when we consider mid-tier firms in columns 5 and 6. The odds ratio associated to a close team member connection is 6.57, while, in contrast, the odds ratio for a distant team member connection is 1.62 and is not statistically different than zero.⁴⁷

We also consider a second measure of within-team interaction based on gender; our focus on gender is motivated by past research that finds individuals are more likely to interact with others sharing similar characteristics (Dustmann et al. 2015; Bandiera, Barankay, and Rasul 2009; Bertrand, Luttmer, and Mullainathan 2000). According to columns 3 and 4, we find team effects are stronger when both the student and the connection share the same gender with

⁴⁷ In unreported results, we estimate whether the effect of a geographically close team-level connection is statistically different from an average team-level connection under a linear probability model. We estimate the difference is statistically significant at the 1% level.

magnitudes similar to those based on residential distance.⁴⁸ Taken together, these results provide strong support to Hypothesis 2b as strong social ties have a greater influence on employment outcomes relative to weaker ties.

6.4 Do social networks affect the recruitment of *high-ability* managers?

6.4.1 Cognitive ability as predicted job performance. Thus far, our results confirm that social networks affect managerial recruitment. While we are the first to demonstrate this result under an experimental setting, our findings are in line with past evidence. However, whether the firm is helped or harmed by these social networks remains unclear. Therefore, we next separately examine whether connected firms are more likely to hire a low-, mid-, or high-ability manager, respectively. We measure predicted job performance by cognitive ability, defined as a manager's score on the quantitative section of the GMAT, and then break these scores into three terciles.

We report the estimated coefficients in Table 6. columns 1 and 2 estimate the effects for top- and mid-tier firms, and find limited evidence that team networks affect the likelihood that a low-ability manager enters the firm. Specifically, for managers with quant GMAT scores in the bottom third of the class, connected managers do not appear to benefit from social networks as the estimated coefficient is economically small and not statistically different than zero. Next, for managers in the mid-tercile of quantitative GMAT scores, social networks have an economically relevant impact as the odds ratio of a connection equals 2.01 and 3.05 for top- and mid-tier firms, respectively. Both coefficients are statistically significant at the 5% level. Finally, for managers in the top-tercile of quantitative GMAT scores, social networks have the largest effect on their employment outcomes at top- and mid-tier firms. We find that the odds ratio of a team connection equals 2.97 and 3.23 for top- and mid-tier firms, respectively. The coefficients are statistically significant at the 1% and 5% levels for top- and mid-tier firms, respectively.

Though the effects of social networks increase with cognitive ability, they also unveil three interesting patterns. First, social networks do not affect the recruitment of low-ability managers, refuting the hypothesis that nepotism prevails in our managerial labor networks. Second, in line with previous results and Hypothesis 2a, the effect of social networks is larger among high- and mid-ability managers in mid-tier firms. Third, the relative impact of social networks on mid- versus high-ability managers increases from top- to mid-tier firms. In top-tier firms, the odds ratio of a connection for high-ability managers is 48% larger than the odds ratio for mid-ability managers. In contrast, in mid-tier firms, this difference is only 6%. Put more directly, social networks support the recruitment of high-ability managers among top-tier firms; among

⁴⁸ In unreported results, we estimate whether the effect of a same-gender team-level connection is statistically different from an average team-level connection under a linear probability model. We estimate the difference is statistically significant at the 1% level.

Table 6
Network effects across high- and low-cognitive-ability graduates

	GMAT quant		Total GMAT		MBA grades	
	Top-tier	Mid-tier	Top-tier	Mid-tier	Top-tier	Mid-tier
Connection x high-ability	1.089*** (3.03)	1.172** (2.11)	1.037*** (2.72)	1.128* (1.96)	0.877*** (2.64)	1.035* (1.93)
Connection x mid-ability	0.700* (1.70)	1.114* (1.67)	0.636* (1.72)	1.374** (2.43)		
Connection x low-ability	0.448 (1.01)	0.752 (1.07)	0.673 (1.48)	0.518 (0.80)	0.688** (2.08)	0.786 (1.58)
Gender FE	Yes	Yes	Yes	Yes	Yes	Yes
Race FE	Yes	Yes	Yes	Yes	Yes	Yes
Nationality FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm × year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2,176	2,176	2,176	2,176	2,007	2,007

This table presents the team-level network results for both high- and low-ability graduates under a discrete choice model. The dependent variable denotes the firm that hired the given manager within 1 year of graduation. Columns 1, 3, and 5 report the network regression results when the employment choices are top-tier firms, while columns 2, 4, and 6 report the network regression results when the employment choices are mid-tier firms. We define the network at the team level and define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular team. Ability is measured based on quant GMAT score (split into terciles) in columns 1 and 2; combined GMAT score (split into terciles) in columns 3 and 4; and first-semester MBA course grades (split at the median) in columns 5 and 6. Controls include fixed effects for graduation year x firm, gender, nationality, and race. Standard errors are computed under the observed information matrix. * $p < .1$; ** $p < .05$; *** $p < .01$.

mid-tier firms, social networks support the recruitment of both mid-ability and high-ability managers.⁴⁹

6.4.2 Alternative measures of cognitive ability. For further confirmation of the results, we consider alternative measures of predicted job performance and present our findings in columns 3 through 6 of Table 6. We focus on Total GMAT scores in columns 3 and 4.⁵⁰ According to columns 3 and 4, a connection among high-ability managers, measured based on combined GMAT score, leads to an odds ratio of 2.82 and 3.09 for top- and mid-tier firms, respectively. In comparison, we find no effect for low-ability managers. Overall, the patterns described in the previous section for columns 1 and 2 prevail for the results reported in columns 3 and 4.

In columns 5 and 6, we instead use predicted job performance based on academic grades in the first semester of the MBA. In the first semester, all MBA students are enrolled in the Core Curriculum. These courses cover eight

⁴⁹ To confirm this interpretation, we estimate whether the effect of connections is statistically different across managerial ability based on a linear probability model. Among top-tier firms, we estimate connections increase the likelihood of hiring a given manager by 0.8%. The effect of a connection increases by 2.5% when the manager is of high-ability (based on quantitative GMAT score) and this difference is statistically significant at the 1% level. Among mid-tier firms, we estimate connections increase the likelihood of hiring a given manager by 0.9%. The effect of a connection increases by 1.3% when the manager is of mid-ability (based on quantitative GMAT score) and this difference is statistically significant at the 1% level.

⁵⁰ Total GMAT scores combine the quantitative and verbal scores.

topics taught by eight different faculty members: Critical Thinking, Economics, Finance, Accounting, Marketing, Operations, Quantitative Analysis, and Strategic Management. While each course is graded separately, we average these grades across all eight courses. We then distinguish between students receiving a B+ grade or above in the core courses, and those with B grade or below.

We note the smaller sample size as we are missing grades for the class of 2006. According to our findings, the odds ratio of a connection for high-ability managers is 2.40 and 2.82 for top- and mid-tier firms, respectively. In comparison, the odds ratio of a connection for low-ability managers is 1.99 and 2.19 for top- and mid-tier firms, respectively. The effects are again significantly larger among managers with higher predicted job performance.

6.4.3 Predicted job performance of connected hires. Next, we estimate differences in the predicted job performance of connected and unconnected hires. Underlying this hypothesis is the implicit assumption that there exist significant differences in ability across teams and these differences affect a firm's ability to recruit through networks. We confirm this requirement in Figure 4. In the right (left) panel, we estimate the mean combined (quantitative) GMAT score for each team in our sample and then plot the distribution of mean scores across teams in the same graduation year. Our data include a total of 566 teams between 2003 and 2013, or over 50 teams in each graduating class. Comparing teams within the same graduating year, we estimate a 30 (3)-point difference in the combined (quantitative) GMAT scores between teams in the 25th percentile and teams in the 75th percentile. Similarly, we estimate a 62 (6)-point difference in scores between the 10th and 90th percentiles. For comparison, the standard deviation is 55 (5.5) points across all managers. Overall, the evidence outlines that cognitive ability is not equally distributed across teams.

With this evidence in hand, we next evaluate whether a manager connected to their employer after graduation holds higher cognitive ability than an unconnected hire in the same employer. As before, we measure ability based on (a) GMAT quantitative score in columns 1 and 2, (b) GMAT combined score in columns 3 and 4, and (c) academic performance in columns 5 and 6. The dependent variable is the score of the connected manager minus the mean score of all managers employed at the same firm. As we are not limited by computational restrictions from the conditional logistic regression, our analysis includes all managers regardless of employer. Our independent variable of interest is a binary variable denoting the hired manager was connected to the employer through a team member. In the odd columns, we control for cohort fixed effects; in the even columns, we also control for characteristics of the manager.

We present our findings in Table 7. According to columns 1 and 2, we estimate a connected manager has a quantitative GMAT score 2–3 points higher than the mean IU MBA graduate hired by the same firm. For comparison, recall

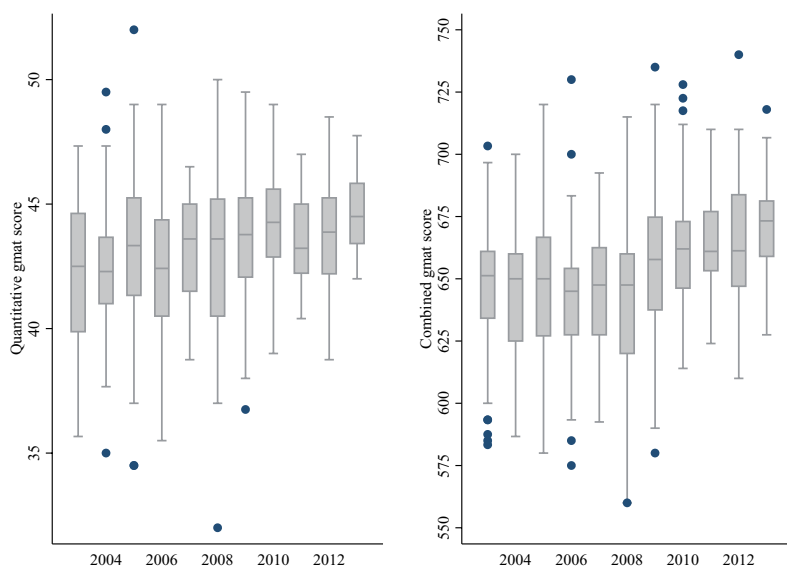


Figure 4
Mean GMAT scores across teams.

This figure plots the mean quant GMAT scores and combined GMAT scores across teams within each graduation year. The left figure present quant GMAT scores, while the right figure present combined GMAT scores. The bottom of the box is the 25th percentile, the inner line is the 50th percentile, and the top of the box is the 75th percentile. The lower extreme is 1.5 times the interquartile range (IQR) below the 25th percentile and the upper extreme is the 1.5 times the interquartile range (IQR) above the 75th percentile.

an average quant GMAT score is 43 points in our sample. In columns 3 and 4, we estimate a connected manager has a 26–28 higher combined GMAT score than an unconnected hire, compared to an average GMAT score of 654 points. In columns 5 and 6, we estimate connected managers receive a 2% higher coursework grades than unconnected managers compared to an average class percentage of 85 points, although these estimates are not statistically different than zero. Across all three measures, we find evidence that unconnected managers hold higher predicted job performance relative to their unconnected peers, showing firms hire managers with higher ability that would not be available in the absence of social networks.

6.4.4 Exploiting predicted job performance across teams. An alternative test to Hypothesis 3 exploits variation in ability across teams rather than individuals. Under our theory of networks, firms randomly connected to high GMAT teams will have access to a pool of talent with higher ability. For illustration, assume that Goldman Sachs and JP Morgan are connected to teams A and B, respectively. Now, let us assume that the average of GMAT scores of team A are higher than the average scores of team B. Since Goldman Sachs is randomly connected to team A, Goldman has a greater likelihood of hiring a high GMAT graduate than JP Morgan.

Table 7
Networks effects on the distribution of cognitive ability

	Quant GMAT score		Combined GMAT score		Class performance	
	(i)	(ii)	(i)	(ii)	(i)	(ii)
Connection	2.785*** (2.65)	2.339** (2.56)	28.124** (2.08)	25.630** (2.13)	2.023 (1.38)	1.898 (1.31)
Prior employee	0.716** (2.24)	0.203 (0.68)	5.508 (1.45)	2.007 (0.55)	0.550* (1.88)	0.437 (1.46)
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Gender FE	No	Yes	No	Yes	No	Yes
Race FE	No	Yes	No	Yes	No	Yes
Nationality FE	No	Yes	No	Yes	No	Yes
N	1,927	1,927	1,927	1,927	1,781	1,781
R-squared	.025	.2	.031	.16	.15	.22

This table presents team-level networks results under linear regressions. This table analyzes student-level observations. In columns 1 and 2, the dependent variable is the quant GMAT score minus the mean quant GMAT of the firm's average hire; in columns 3 and 4, the dependent variable is the mean combined GMAT score minus the mean combined GMAT of the firm's average hire; and in columns 5 and 6 the dependent variable is first semester course grades minus the mean course grades of the firm's average hire. Connection is a binary variable denoting the graduate was connected to their first post-MBA employer through a team member. In all specifications, we control for whether the hire was previously employed with the firm and include fixed effects for graduation year, gender, nationality, and race. We cluster all results at the level of the team level. * $p < .1$; ** $p < .05$; *** $p < .01$.

We provide a formal test of this hypothesis in Table A5 of the Internet Appendix. Across all three measures of ability (quantitative GMAT score, combined GMAT score, or MBA course percentage grade), we estimate that a 10-point increase in the mean ability of the team leads to a 1- to 3-point increase in the score of the new hire (when compared to the mean score of all hires at the same firm). All results are statistically significant at the 5% level. The results again suggest networks affect the likelihood of hiring a manager with high predicted job performance.

6.4.5 Tenure of connected hires. According to our results, social connections increase the likelihood a firm hires a given talented manager, with limited effects on the likelihood of hiring a less-talented manager. However, for firms to substantially benefit from these social networks, managers must also remain employed with the firm. Therefore, we next estimate how connections affect tenure at the firm. We report our findings in Table A6 of the Internet Appendix. The dependent variable is a binary outcome identifying whether the manager remained remained with their first post-MBA employer longer than the mean hire of that same firm. We estimate connected managers are 26%–27% more likely to stay with their first employer relative to unconnected managers. Our results validate prior evidence, including Dustmann et al. (2015); Brown, Setren, and Topa (2016) that social connections reduce mismatch between applicants and firms. As a result, connections lead to long-term benefits for the firm.

6.5 Why do social networks affect the recruitment of managers?

6.5.1 Survey of recent and current Indiana University MBA students.

The results above offer clear evidence that social networks influence the employment outcomes of managers; however, the precise channel is less clear. We provide one argument: social networks help managers may learn about firm fundamentals (Grossman and Hart 1983; Hart 1983; Azariadis 1983). To provide evidence for this argument, we next survey current Indiana University MBA students from the graduating classes of 2020 and 2021. We contacted students in the spring of 2020 following team interactions. There are a total of 194 and 134 MBA students in the class of 2020 and 2021; we received a total of exactly 80 responses from each class, leading to a response rate of 41% and 60% for the 2020 and 2021 class, respectively. We then match this information with three data sources: (1) admissions information detailing race, nationality, and gender, (2) employment histories collected from the online business networking service, and (3) team assignments. By merging this information with self-reported survey data, we can identify the effect of social networks on the behavior and views of the applicants. We provide summary statistics for these two samples in panel A of Table 8. In Table A4 of the Internet Appendix, we confirm survey respondents are representative of the full class, as we do not find differences between respondents and nonrespondents based on demographics and MBA intended majors.

6.5.2 Applying to a connected firm. As discussed above, we hypothesize social networks may influence the job preferences of managers. To confirm this interpretation of the results, we next evaluate how team networks influence job application decisions. Therefore, we ask students in the class of 2020 and 2021, “Did you apply to positions with any of the firms below? If yes, please check all firms you applied to. If no, please check None.” For students in the first year of the MBA (class of 2021), we ask about applications to summer internships and for second-year students (in the class of 2020), we ask about full-time employment after graduation. We limit the options to the 79 employers in the sample that (a) employed at least one IU MBA student from the class of 2020 or 2021 prior to joining the MBA and (b) recruit multiple graduates from the three top-ranked MBA programs. The exact list of firms is reported in Table A13 of the Internet Appendix, and is similar to the combined set of firms in top- and mid-tier groups, as reported in Table A1 and A2 of the Internet Appendix.⁵¹

According to panel A of Table 8 we estimate the median (median) student applies to 7.1 (6) of the 79 firms in our sample. The results offer confirmation of Kuhnen (2017), who find that UNC MBA students submit 13 applications on average across all employers. Overall, we find students apply to a select

⁵¹ There are small differences between the two lists of firms since some top- and mid-tier firms no longer exist or did not exist in the earlier sample.

Table 8
Summary of survey responses and network effects for 2020 and 2021 classes

<i>A. Summary of 2020 and 2021 Indiana University MBA students</i>				
	N	Mean	Median	Std
Mid- or top-tier firm after MBA	160	0.275	0.000	0.448
Mid- or top-tier firm before MBA	160	0.169	0.000	0.376
Number of applications	160	7.100	6.000	5.696
Returning to a prior employer	160	0.062	0.000	0.243
Accept internship offer	80	0.425	0.000	0.497
Received a referral from classmate	160	0.144	0.000	0.352
Number of team connections	160	0.644	1.000	0.721
Team size	160	4.981	5.000	1.481
Graduation year	160	2,020.475	2,020.000	0.501
<i>B. Network effects on applications</i>				
Submitted application				
Connection	0.191** (2.54)			0.192** (2.57)
Cohort FE	Yes			No
Firm x year FE	Yes			Yes
N	4,977			5,056
R-squared	.29			.29
<i>C. Network effects on total application rate and referrals</i>				
	Number of applications		Received referral	
	(i)	(ii)	(i)	(ii)
Connection	0.091 (0.14)	0.387 (0.69)	0.024 (0.58)	0.023 (0.56)
Prior employee	0.758 (0.59)	0.848 (0.76)	0.013 (0.17)	0.019 (0.26)
Year FE	Yes	Yes	Yes	Yes
Gender FE	No	Yes	No	Yes
Race FE	No	Yes	No	Yes
Nationality FE	No	Yes	No	Yes
N	160	160	160	160
R-squared	.0056	.17	.0038	.037

This table reports the survey responses of students from the IU MBA Classes of 2020 and 2021. Panel A summarizes the responses across all students. The variable Accept Internship Offer has a smaller sample as it is only measured for the class of 2020. Panel B reports the network regression results for the Classes of 2020 and 2021 under a linear probability regression. Responses are aggregated to the team level. The dependent variable denotes the number of team members that submitted an application to the firm relative to the average application rate to the firm. The choice of firms includes both top- and mid-tier firms as defined in the Internet Appendix. Connection is a binary variable denoting the team is connected to the given firm through a team member. Controls include cohort fixed effects and firm x year fixed effects. Errors are clustered separately at the year level and at the firm level. Panel C reports additional survey results for the classes of 2020–2021. In columns 1 and 2, the dependent variable is a discrete variable denoting the number of job applications submitted by a graduate. In columns 3 and 4, the dependent variable is a binary variable denoting whether the graduate received a referral from any classmate. Connection is a binary variable that takes a value of one when the graduate is connected to any of top- or mid-tier firm through a team member. Prior Employee is a binary variable denoting the graduate worked any top- or mid-tier firm prior to the MBA. We cluster all results at the team level. * $p < .1$; ** $p < .05$; *** $p < .01$.

number of firms, which substantially restricts an employer’s access to high-ability managers. Next, we evaluate whether connections to a given firm at the team level increase the likelihood of applying to that firm. For each possible firm-manager pair, we measure whether given manager i applied to given

firm j . We then aggregate the results to the team-level excluding the prior employee, similar to the specification in panel B of Table 4. Following the aggregate specification in Table 4, we cluster errors at both the firm and the year levels. We present our findings in panel B of Table 8. We estimate a team connection to a given firm increases the likelihood of applying to the firm by 19 percentage points and the effect is statistically significant at the 1% level. These results highlight the importance that former employees have on influencing the application process of their peers, supporting the view that information is likely flowing from former employees to applicants.

6.5.3 Applying to other firms. Combined, the survey results suggest that social networks increase the likelihood of applying to the connected firm. However, it is unclear whether the connections also drive the application rate to outside firms. Therefore, in panel C of Table 8, we evaluate whether students connected to the 79 firms included in our analysis apply to more firms in general. In the first column, we control only for the year of graduation, while in the second column we also control for student demographics. We find no evidence of any increase in the total application rate, suggesting the application effect is specific to the connected firm.

6.5.4 Incentives to provide information. Given the above results, one might question the incentives that former employees have to provide information about their past employers. If former employees plan on returning to their former employers and their jobs are not yet secured, they might be discouraged to share information as they might want to reduce additional competition. However, if they plan on switching careers or if they attended the MBA program under a job contract that requires them to return to their former employer, then they may provide information to peers as they may seek future reciprocity from their peers. Using our survey responses, we find that only 6% of the respondents plan to return to their former employer. For comparison, we also estimate only 4.5% of the 2,176 graduates from the classes of 2003–2013 return to a former employer within the first 5 years after graduation. This is a rather low percentage, consistent with the premise that most MBA students aspire a career switch, and explaining why most students in our setting are willing to share information about their former employer.

6.5.5 Referrals. The survey evidence presented above suggests connections lead potential managers to prefer the firm over alternative employers. However, this does not necessarily imply other channels are not relevant. For instance, past empirical research in labor economics argues that social benefits instead benefit firms primarily by offering referrals of applicants. There are two common arguments regarding the role of referrals. In the first version, valuable information is implicit in social networks because current and past employees are of known high-ability and interact with workers of similar

quality (Montgomery 1991; Hensvik and Skans 2016). In the second version, information is valuable because current and past workers have explicit or implicit incentives to be truthful when actively providing referrals (Pallais and Sands 2016; Heath 2018). Firms will then disproportionately make offers to candidates with referrals. In both versions of these models, we should continue to find that social networks increase the likelihood an applicant joins the employer. Additionally, if cognitive ability is not readily observable to firms, but is observable to social networks, referrals may provide critical information about cognitive ability. As a result, connected managers will be of higher cognitive ability than unconnected managers.

Given the potential role of referrals in supporting our findings, we next evaluate the role of former employees in providing firms information about the managers in our setting. According to panel A of Table 8, we find that 14% of students report receiving a referral from another classmate. In addition, we test whether the sample of firms included in our analysis, high- and mid-tier firms, rely less on referrals than other employers. According to panel C of Table 8, we find that connections to top- and mid-tier firms have a similar effect on the number of referrals received. Overall, the results suggest a potential role for referrals in our analysis.

6.6 Extensions to discrete-choice analysis

6.6.1 Independence of irrelevant alternatives. Discrete choice models require the economist to make several decisions in the estimation process. One particular decision is whether to assume Independence of Irrelevant Alternatives (IIA). In discrete choice theory, the IIA assumption says that when people are asked to choose among a set of alternatives, their odds of choosing A over B should not depend on whether some other alternative C is present or absent. An alternative way to state IIA is that any item added to the set of choice will decrease all other items' likelihood by an equal fraction.

Applied to our setting, the particular concern is that graduates likely first choose to apply to a set of industries, and then consider firms within these industries. In this instance, adding a potential employer in industry k to the choice model will then affect the recruitment of other firms in industry k , but have a more limited effect on firms in other industries. To overcome this issue, we implement a nested logistic regression that assumes graduates (a) first choose an industry and then (b) choose an employer within the industry; we sort firms into industries as defined by two-digit NAICS codes.

By introducing this specification, employment preferences can now be correlated within each industry. The nested logit model allows for the violation of the IIA-assumption by imposing that the error terms of alternatives within a nest can be correlated while the error terms of alternatives across nests are uncorrelated. As a result, we can estimate a nested logit as a combination of multiple standard logit models. In our updated specification, we model the random utility of manager i joining firm j in industry m as $U_{i,j,m} =$

Table 9
Network effects under a nested-logit specification

	Team				Cohort	
	(i)	(ii)	(i)	(ii)	(i)	(ii)
Team connection	0.836*** (2.64)	1.011*** (2.81)				
Cohort connection					0.452* (1.94)	0.580*** (2.82)
Team connection x High-ability			1.098** (2.34)	1.083** (2.29)		
Team connection x Mid-ability			0.671 (1.34)	0.702 (1.36)		
Team connection x Low-ability			0.700 (1.13)	0.695 (1.11)		
Gender FE	No	Yes	No	Yes	No	Yes
Race FE	No	Yes	No	Yes	No	Yes
Nationality FE	No	Yes	No	Yes	No	Yes
GMAT FE	No	Yes	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	2,176	2,176	2,176	2,176	2,176	2,176

This table presents the network results under a nested logit model. We implement a two-stage nested model where graduates first choose an industry for employment and then choose employment among the firms in that industry. The dependent variable denotes the top-tier firm hired the given manager within 1 year of graduation. Columns 1–4 define network at the team level and define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular team. Columns 3 and 4 split graduates into terciles based on cognitive ability as measured by quantitative GMAT scores. Columns 5 and 6 define network at the cohort-level and define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular cohort. Controls include fixed effects for graduation year x firm, gender, nationality, GMAT score (split at the median), and race. Standard errors are computed under the observed information matrix. * $p < .1$; ** $p < .05$; *** $p < .01$.

$U_{i,m} + U_{i,j|m} = (\eta_{i,m} + \varepsilon_{i,m}) + (\eta_{i,j|m} + \varepsilon_{i,j|m})$. The errors terms $\varepsilon_{i,m}$ and $\varepsilon_{i,j|m}$ are independent of each other. For each manager i , error terms $\varepsilon_{i,j|m}$ are identically and independently distributed (iid) extreme-value.

We present our findings under the nest logit specification in Table 9. The first and second columns report the effect of a team-level connection across the mean manager. The third and fourth columns distinguish managers by cognitive ability based on the quantitative GMAT score. The fifth and sixth columns report of effect of a cohort-level connection. Across all specifications, the results remain unchanged relative to those estimated with a conditional logit, suggesting that the IIA in our case might not be an unreasonable assumption.

6.6.2 Placebo networks. A second concern is that cohorts and teams are assigned by balancing student characteristics; therefore, our estimated effects may be driven by balancing rather than networks. We believe this is unlikely for two reasons. First, we find no evidence that controlling for characteristics included in the balancing criteria has any effect on our estimates. Second, given teams compose only four managers, it is not possible to balance firm or industry connections across teams.

For further evidence balancing is not driving the results, we place students into placebo teams in each year and then estimate whether placebo connections

influence recruiting outcomes. For this analysis, we return to our conditional logistic regression specification. To confirm the results are not driven by the team assignment criteria, students are placed into placebo teams that retain the demographics of teams including gender, ethnicity, and nationality. We present our findings in Table 10. Panel A focuses on top-tier firms, while panel B focuses on mid-tier firms. We estimate network effects at employment within 1 year of graduation and 5 years of graduation. We also estimate networks effects for the mean graduate and separately for each tercile of cognitive ability. We find no evidence that placebo team members affect student employment outcomes. We take the lack of results as evidence that teams are acting as social networks.

6.6.3 Balancing across industries. A third concern is that connections to a particular industry correlate with variables that predict employment. Based on this concern, we evaluate whether connections to a given industry affect the likelihood of employment within that industry (absent the connected firm). According to the results in Table 11, we find no evidence that industry-level connections influence employment with other firms in the industry.

6.7 Rejecting alternative mechanisms

We conclude this section by rejecting alternative channels that may explain our results. We consider two possible alternatives. First, social connections might help employees gain employment in any top-tier firm rather than the connected firm. Second, peers with past experience in top-tier firms may improve classroom performance, which in turn, will affect employment outcomes.

6.7.1 Job application and interviewing skills. The firms included in our baseline analysis each hired multiple graduates from top MBA programs during our sample period. Given the competition to gain employment with these firms, the IU MBA graduates in our sample may depend on peers to gain information about the job application and interviewing process. Graduates connected to these firms then may be more likely to join these firms following graduation because of improved skills. This likelihood may be driving our network results.

However, assuming knowledge about the job application and interview process is likely valuable across all firms, rather than exclusively the connected firm, we should expect connections to one firm will influence the likelihood of employment across outside firms as well. We test this interpretation of our findings in the first and second columns of Table 12. The independent variable is a binary variable denoting whether a given graduate is connected to any top-tier firm through a team member.⁵² As the outcome variable is employment with any of the 43 firms included in our conditional logistic analysis, the dependent variable is a simple binary outcome. We run this specification as

⁵² In unreported results, we include both top- and mid-tier firms. The results are similar.

Table 10
Placebo network effects

<i>A. Placebo network effects on top-tier firms</i>					
	1 year			5 years	
Connection	−0.182 (−0.78)	−0.217 (−0.92)		−0.052 (−0.21)	−0.039 (−0.16)
Connection x High-ability			0.159 (0.43)		
Connection x Mid-ability			−0.196 (−0.48)		
Connection x Low-ability			−0.387 (−0.81)		
Gender FE	No	Yes	Yes	No	Yes
Race FE	No	Yes	Yes	No	Yes
Nationality FE	No	Yes	Yes	No	Yes
GMAT FE	No	Yes	Yes	No	Yes
Firm × year FE	Yes	Yes	Yes	Yes	Yes
N	2,176	2,176	2,176	2,176	2,176
<i>B. Placebo network effects on mid-tier firms</i>					
	1 year			5 years	
Connection	−0.150 (−0.47)	−0.837 (−1.50)		0.009 (0.02)	−0.147 (−0.40)
Connection x High-ability			0.129 (0.26)		
Connection x Mid-ability			−0.748 (−1.43)		
Connection x Low-ability			−0.357 (−0.51)		
Gender FE	No	Yes	Yes	No	Yes
Race FE	No	Yes	Yes	No	Yes
Nationality FE	No	Yes	Yes	No	Yes
GMAT FE	No	Yes	Yes	No	Yes
Firm × year FE	Yes	Yes	Yes	Yes	Yes
N	2,176	2,176	2,176	2,176	2,176

This table presents the placebo team-level network results under a discrete choice model. Placebo teams are created by randomly assigning students, while keeping the distribution of team characteristics similar to the distribution across actual teams. The dependent variable denotes the firm that hired the given manager. Panel A considers employment options across high-tier firms, while panel B considers employment options across mid-tier firms. Columns 1–3 measure employment 1 year after graduation; columns 4 and 5 measure employment 5 years after graduation. We define the network at the team level and define Connection as a binary variable that takes a value of one when the firm previously employed a student in that particular team. Column 5 splits graduates into terciles based on quantitative GMAT score. Controls include fixed effects for graduation year x firm, gender, nationality, GMAT score (split at the median), and race. Standard errors are computed under the observed information matrix. * $p < .1$; ** $p < .05$; *** $p < .01$.

a linear probability model. Column 1 controls for year fixed effects, while column 2 also controls for graduate demographics. Across both specifications, we estimate a connection has no impact on employment opportunities outside the connected firm. We take these results as evidence that managers are not learning about the job application and interviewing process through networks.

6.7.2 Classroom learning. As top-tier firms routinely employ graduates from high-ranking MBA programs, IU students with prior experience at these firms

Table 11
Network effects on industry choice

	Other top-tier firms in same industry		Other mid-tier firms in same industry	
	Baseline	Ability	Baseline	Ability
Connection	-0.369 (-1.56)		-0.545 (-1.33)	
Connection x High-ability		-0.432 (-1.01)		0.344 (0.71)
Connection x Mid-ability		-0.737 (-1.60)		-1.136 (-1.28)
Connection x Low-ability		-0.023 (-0.07)		-1.971* (-1.69)
Gender FE	Yes	Yes	Yes	Yes
Race FE	Yes	Yes	Yes	Yes
Nationality FE	Yes	Yes	Yes	Yes
GMAT FE	Yes	Yes	Yes	Yes
Firm × year FE	Yes	Yes	Yes	Yes
N	2,176	2,176	2,176	2,176

This table presents the team-level network results across industries under a discrete choice model. The dependent variable denotes the firm that hired the given manager. Columns 1 and 2 report the network regression results when the employment choices are top-tier firms, while columns 3 and 4 report the network regression results when the employment choices are mid-tier firms. Connection is a binary variable that takes a value of one when the graduate is connected to a different firm in that industry through a team member. Columns 2 and 4 measure cognitive ability based on quantitative GMAT scores. Controls include fixed effects for graduation year x firm, nationality, GMAT score (split at the median), and race. Standard errors are computed under the observed information matrix. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 12
Alternative explanations

	Employment with any top-tier firm		Classroom performance	
	(i)	(ii)	(i)	(ii)
Connection	0.009 (0.46)	0.008 (0.41)	0.029 (0.14)	0.077 (0.40)
Prior employee	0.073*** (3.06)	0.072*** (3.01)	0.710*** (3.23)	0.768*** (3.61)
Cohort FE	Yes	Yes	Yes	Yes
Gender FE	No	Yes	No	Yes
Race FE	No	Yes	No	Yes
Nationality FE	No	Yes	No	Yes
N	2,175	2,175	2,007	2,007
R-squared	.024	.026	.23	.31

This table reports alternative explanations of the network results. In columns 1 and 2, the dependent variable is a binary variable that takes a value of one when the graduate joins any top-tier firm following graduation. In columns 3 and 4, the dependent variable is a continuous variable measuring the first-semester course grades of the graduate. Connection is a binary variable that takes a value of one when the graduate is connected to any top-tier firm through a team member. Prior Employee is a binary variable denoting the graduate worked for any top- or mid-tier firm prior to the MBA. Across both panels, controls include fixed effects for year of graduation, gender, race, and nationality. We cluster all results at the team level. * $p < .1$; ** $p < .05$; *** $p < .01$.

may hold high academic abilities relative to other IU MBA students. As a result, these students may have a positive influence on the first semester grades of team members, ultimately affecting the employment opportunities of their peers. We test this interpretation of our findings in the third and fourth columns of Table 12.

The outcome variable is now the first semester grades of each student; column 3 includes year fixed effects; and column 4 includes demographics fixed effects as well. We find no evidence team members with prior employment at top- or mid-tier firms influence the grades of their teammates.

7. Conclusion

This paper evaluates whether social networks help or hinder firms competing for talented young managers. Relying on the random assignment of MBA students into teams and cohorts, we can precisely isolate peer effects on prospective employers; by incorporating MBA admissions data with online profiles, we are able to follow individual managers over their careers and develop measures of predicted job performance. We find connections disproportionately affect the recruitment of high-ability managers with no effects for low-ability managers. In addition, these effects are larger for tight connections and more valuable for geographically distant firms. By exploiting variation in cognitive ability across teams, we find well-connected firms hire more capable managers. Finally, by directly surveying managers, we find networks encourage managers of all abilities to apply for new positions, leading to greater access to talent.

To our knowledge, we offer the first evidence that social networks improve managerial recruiting efforts. We recognize that in contrast to past research that focuses explicitly on CEOs, we analyze the market for mid-level managers: nearly 60% of our sample hold a managerial title within 1 year of graduation and over 70% within 5 years of graduation. Despite the differences, we believe our analysis of the young managerial labor market holds valuable lessons for the labor market of more senior managers. First, while the graduates in our sample are not yet CEOs, many will eventually reach that status: among public firm CEOs, Indiana University is 1 of the 10 most common programs to graduate from (Bhagat, Bolton, and Subramanian 2010). Second, as the vast majority of firms in the United States are quite small—over 89% of firms have under 20 employees, 9% have 20–99 employees and 1.5% have 100–499 employees—many firms likely face constraints when recruiting managers due to informational frictions and insufficient resources. Third, recent evidence finds that even the labor market for public firm CEOs is actually quite segmented; for example, 30% of public firm CEOs were born in the same state as the firm's headquarter, a rate five times the expected rate if labor markets were national (Yonker 2016). Because of this segmentation, social networks are likely to continue to influence the decision to apply for positions and accept offers, even among the top management team. Confirming this interpretation is a valuable avenue for future research.

References

- Abowd, J. M., F. Kramarz, and D. N. Margolis. 1999. High wage workers and high wage firms. *Econometrica* 67:251–333.

- Agrawal, A. K., I. Hacamo, and Z. Hu. 2020. Information dispersion across employees and stock returns. *Review of Financial Studies*. Advance Access published September 16, 2020, 10.1093/rfs/hhaa105.
- Ahern, K. R., R. Duchin, and T. Shumway. 2014. Peer effects in risk aversion and trust. *Review of Financial Studies* 27:3213–40.
- Arcidiacono, P., J. Cooley, and A. Hussey. 2008. The economic returns to an MBA. *International Economic Review* 49:873–99.
- Azariadis, C. 1983. Employment with asymmetric information. *Quarterly Journal of Economics* 98:157–72.
- Bandiera, O., I. Barankay, and I. Rasul. 2009. Social connections and incentives in the workplace: Evidence from personnel data. *Econometrica* 77:1047–94.
- Beaman, L., and J. Magruder. 2012. Who gets the job referral? evidence from a social networks experiment. *American Economic Review* 102:3574–93.
- Belo, F., J. Li, X. Lin, and X. Zhao. 2017. Labor-force heterogeneity and asset prices: The importance of skilled labor. *Review of Financial Studies* 30:3669–709.
- Bennedsen, M., F. Pérez-González, and D. Wolfenzon. 2016. Do ceos matter? Evidence from hospitalization events. *Journal of Finance* 75:1877–911.
- Bertrand, M., and A. Schoar. 2003. Managing with style: The effect of managers on firm policies. *Quarterly Journal of Economics* 118:1169–208.
- Bertrand, M., E. F. P. Luttmer, and S. Mullainathan. 2000. Network effects and welfare cultures. *Quarterly Journal of Economics* 115:1019.
- Bertrand, M., C. Goldin, and L. F. Katz. 2010. Dynamics of the gender gap for young professionals in the financial and corporate sectors. *American Economic Journal: Applied Economics* 2:228–55.
- Bhagat, S., B. J. Bolton, and A. Subramanian. 2014. Ceo education, ceo turnover, and firm performance. Working, University of Colorado at Boulder.
- Bhole, M., and P. Oyer. 2014. Do MBAs pick winning stocks when choosing their first job? Working Paper, Stanford University.
- Brown, M., E. Setren, and G. Topa. 2016. Do informal referrals lead to better matches? Evidence from a firm's employee referral system. *Journal of Labor Economics* 34:161–209.
- Burks, S. V., B. Cowgill, M. Hoffman, and M. Housman. 2015. The value of hiring through employee referrals. *Quarterly Journal of Economics* 130:805–39.
- Cai, Y., and M. Sevilir. 2012. Board connections and m&a transactions. *Journal of Financial Economics* 103:327–49.
- Carrell, S. E., R. L. Fullerton, and J. E. West. 2009. Does your cohort matter? Measuring peer effects in college achievement. *Journal of Labor Economics* 27:439–64.
- Chang, Y. Y., S. Dasgupta, and G. Hilary. 2010. Ceo ability, pay, and firm performance. *Management Science* 56:1633–52.
- Chevalier, J., and G. Ellison. 1997. Risk taking by mutual funds as a response to incentives. *Journal of Political Economy* 105:1167–200.
- Custódio, C., M. A. Ferreira, and P. Matos. 2013. Generalists versus specialists: Lifetime work experience and chief executive officer pay. *Journal of Financial Economics* 108:471–92.
- Cziraki, P., and D. Jenter. 2020. The market for ceos. Working Paper, University of Toronto.
- Donangelo, A. 2014. Labor mobility: Implications for asset pricing. *Journal of Finance* 69:1321–46.
- Dustmann, C., A. Glitz, U. Schönberg, and H. Brücker. 2015. Referral-based job search networks. *Review of Economic Studies* 83:514–46.

- Eisfeldt, A. L. and D. Papanikolaou. 2013. Organization capital and the cross-section of expected returns. *Journal of Finance* 68:1365–406.
- Engelberg, J., P. Gao, and C. A. Parsons. Friends with money. 2012. *Journal of Financial Economics* 103:169–88.
- Faccio, M. 2016. Politically connected firms. *American Economic Review* 96:369–86.
- Falato, A., D. Li, and T. Milbourn. 2015. Which skills matter in the market for ceos? Evidence from pay for ceo credentials. *Management Science* 61:2845–69.
- Fracassi, C., and G. Tate. 2012. External networking and internal firm governance. *Journal of Finance* 67:153–94.
- Frydman, C. 2005. Rising through the ranks. the evolution of the market for corporate executives, 1936–2003. Working Paper, Columbia University.
- . 2019. Rising through the ranks: The evolution of the market for corporate executives, 1936–2003. *Management Science* 65:4951–79.
- Gee, K. 2017. Another thing amazon is disrupting: Business-school recruiting. *Wall Street Journal*, October 4.
- Gee, L. K., J. Jones, and M. Burke. 2017. Social networks and labor markets: How strong ties relate to job finding on facebook's social network. *Journal of Labor Economics* 35:485–518.
- Goldberg, M. S. 1982. Discrimination, nepotism, and long-run wage differentials. *Quarterly Journal of Economics* 97:307–19.
- . 1995. Product differentiation and oligopoly in international markets: The case of the us automobile industry. *Econometrica* 63:891–951.
- Gompers, P., J. Ishii, and A. Metrick. 2003. Corporate governance and equity prices. *Quarterly Journal of Economics* 118:107–55.
- Graham, J. R., D. Kim, and H. Kim. 2020. Executive mobility in the united states, 1920 to 2011. Working Paper, Duke University.
- Graham, J. R., H. Kim, S. Li, and J. Qiu. 2013. Human capital loss in corporate bankruptcy. Working Paper, Duke University.
- Graham, J. R., D. Kim, and H. Kim. 2020. Executive mobility in the united states, 1920 to 2011. Working Paper, Duke University.
- Granovetter, M. S. 1997. The strength of weak ties. In *Social networks*, 347–67. Amsterdam, the Netherlands: Elsevier.
- Groen, J. A. 2004. The effect of college location on migration of college-educated labor. *Journal of Econometrics* 121:125–42.
- Grossman, S. J., and O. D. Hart. 1983. Implicit contracts under asymmetric information. *Quarterly Journal of Economics* 98:123–56.
- Hart, O. D. 1983. Optimal labour contracts under asymmetric information: An introduction. *Review of Economic Studies* 50:3–35.
- R. F. Haselmann, D. Schoenherr, and V. Vig. 2018. Rent-seeking in elite networks. *Journal of Political Economy* 126:1638–90.
- Hausman, J., and D. McFadden. 1984. Specification tests for the multinomial logit model. *Econometrica* 52:1219–40.
- Heath, R. 2018. Why do firms hire using referrals? evidence from bangladeshi garment factories. *Journal of Political Economy* 126:1691–746
- Hellerstein, J. K., M. J. Kutzbach, and D. Neumark. 2019. Labor market networks and recovery from mass layoffs before, during, and after the great recession. *Journal of Urban Economics* 113:103192.

- Hellmann, T., L. Lindsey, and M. Puri. 2008. Building relationships early: Banks in venture capital. *Review of Financial Studies* 21:513–41.
- Hensvik, L., and O. N. Skans. 2016. Social networks, employee selection, and labor market outcomes. *Journal of Labor Economics* 34:825–67.
- Hunter, J. E. 1986. Cognitive ability, cognitive aptitudes, job knowledge, and job performance. *Journal of Vocational Behavior* 29:340–362.
- İmrohoroglu, A., and C. Tüzel. 2014. Firm-level productivity, risk, and return. *Management Science* 60:2073–90.
- Jacobson, L. S., R. J. LaLonde, and D. G. Sullivan. 1993. Earnings losses of displaced workers. *American Economic Review* 83:685–709.
- Jensen, M. C., and K. J. Murphy. 1990. Performance pay and top-management incentives. *Journal of Political Economy* 98:225–64.
- Jenter, D., E. Matveyev, and L. Roth. 2020. Good and bad ceos. Working Paper, London School of Economics.
- Kaniel, R., C. Massey, and D. T. Robinson. 2010. The importance of being an optimist: Evidence from labor markets. Working Paper, University of Rochester.
- Karolyi, S. A. 2014. Personal lending relationships. *Journal of Finance* 73:5–49.
- Kennan, J., and J. R. Walker. 2011. The effect of expected income on individual migration decisions. *Econometrica* 79:211–51.
- T. King, A. Srivastav, and J. Williams. 2016. What's in an education? implications of ceo education for bank performance. *Journal of Corporate Finance* 37:287–308.
- Kramarz, F., and O. N. Skans. 2014. When strong ties are strong: Networks and youth labour market entry. *Review of Economic Studies* 81:1164–200.
- Kramarz, F., and D. Thesmar. 2013. Social networks in the boardroom. *Journal of the European Economic Association* 11:780–807.
- Kuhnen, C. M. 2017. Searching for jobs: Evidence from MBA graduates. Working Paper, University of North Carolina, Chapel-Hill.
- Kuhnen, C. M., and P. Oyer. 2016. Exploration for Human Capital: Evidence from the MBA Labor Market. *Journal of Labor Economics* 34:255–86.
- Lerner, J., and U. Malmendier. 2013. With a little help from my (random) friends: Success and failure in post-business school entrepreneurship. *Review of Financial Studies* 26:2411–52.
- Lindsey, L. 2008. Blurring firm boundaries: The role of venture capital in strategic alliances. *Journal of Finance* 63:1137–68.
- Lustig, H., C. Syverson, and S. Van Nieuwerburgh. 2011. Technological change and the growing inequality in managerial compensation. *Journal of Financial Economics* 99:601–27.
- Lyle, D. S. 2009. The effects of peer group heterogeneity on the production of human capital at west point. *American Economic Journal: Applied Economics* 1:69–84.
- Ma, M., J. Pan, and S. R. Stubben. 2020. The effect of local tournament incentives on firms' performance, risk-taking decisions, and financial reporting decisions. *Accounting Review* 95:283–309.
- Manski, C. F. 1993. Identification of endogenous social effects: The reflection problem. *Review of Economic Studies* 60:531–42.
- McFadden, D. L. 1973. Conditional logit analysis of qualitative choice behavior. In *Frontiers in econometrics*, eds. P. Zarembka, 105–42. New York: Academic Press.
- Montgomery, J. D. 1991. Social networks and labor-market outcomes: Toward an economic analysis. *American Economic Review* 81:1408–18.

- Murphy, K. J., and J. Zabojsnik. 2007. Managerial capital and the market for ceos. Working Paper, University of Southern California.
- Nickerson, J. 2017. Market forces and ceo pay: Shocks to ceo demand induced by ipo waves. *Review of Financial Studies* 30:2272–2312.
- Oyer, P. 2008. The making of an investment banker: Stock market shocks, career choice, and lifetime income. *Journal of Finance* 63:2601–28.
- Pallai, A., and E. G. Sands. 2016. Why the referential treatment? evidence from field experiments on referrals. *Journal of Political Economy* 124:1793–828.
- Ree, M. J., and J. A. Earles. 1992. Intelligence is the best predictor of job performance. *Current Directions in Psychological Science* 1:86–9.
- Sacerdote, B. 2001. Peer effects with random assignment: Results for dartmouth roommates. *Quarterly Journal of Economics* 116:681–704.
- Sapienza, P., L. Zingales, and D. Maestripieri. 2009. Gender differences in financial risk aversion and career choices are affected by testosterone. *Proceedings of the National Academy of Sciences* 106:15268–73.
- Schmidt, F. L., and J. Hunter. 2004. General mental ability in the world of work: Occupational attainment and job performance. *Journal of Personality and Social Psychology* 86:162–73.
- . 1998. The validity and utility of selection methods in personnel psychology: Practical and theoretical implications of 85 years of research findings. *Psychological Bulletin* 124:262–74.
- Shue, K. 2013. Executive networks and firm policies: Evidence from the random assignment of MBA peers. *Review of Financial Studies* 26:1401–42.
- Train, K. E. 2009. *Discrete choice methods with simulation*. Cambridge, UK: Cambridge University Press.
- Weisbach, M. S. 1988. Outside directors and ceo turnover. *Journal of Financial Economics* 20:431–60.
- White, H. 1982. Maximum likelihood estimation of misspecified models. *Econometrica* 50:1–25.
- Yonker, S. E. 2016. Geography and the market for ceos. *Management Science* 63:609–30.
- Zhao, H. 2018. Executive labor market segmentation: How local market density affects incentives and performance. *Journal of Corporate Finance* 50:1–21.
- Zimmerman, D. J. 2003. Peer effects in academic outcomes: Evidence from a natural experiment. *Review of Economics and Statistics* 85:9–23.
- Zimmerman, S. D. 2016. Making the one percent: The role of elite universities and elite peers. Working Paper, Institute of Labor Economics.
- Zingales, L. 2000. In search of new foundations. *Journal of Finance* 55:1623–53.